

Humanoids

Humanoid Horizons: Moravec's Paradox

We see good improvement in the World Humanoid Robot Games, but the industry's key focus area is whether it can perform real tasks efficiently and productively. Recent model updates showed an early signal of scaling law. Commercialization progress is the key focus area for investors.

"It is comparatively easy to make computers exhibit adult level performance on intelligence tests, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility." Hans Moravec in 1988. Industry has gradually solved the robot's locomotion. How about perception and physical world reasoning?

Highlights in August:

- World Humanoid Robot Games 2026:** The 2026 Games gathered 666 teams and 2,056 robots (vs. ~280 teams in 2025). Physical capability gains impressed, with several humanoid sports records broken by wide margins (Tiangong ran 100m in 8.64s vs. Usain Bolt's 9.58s and Tiangong's 21.5s in 2025). In applications, compared with 2025, the 2026 Games put much greater weight on multi-step tasks in real-world scenarios vs. single predefined motions. The 21 scenario competitions spanned factories, homes, hotels, retail, hospitals, and emergency response. In catering, for example, robots had to interpret a spoken instruction, retrieve and heat food, collect a drink, and deliver the order. Autonomy was explicitly rewarded, with fully autonomous execution receiving twice the scoring weight of remote operation in relevant scenarios. The Games also introduced a dedicated dexterous-hand competition with eight fine-manipulation tasks, including weighing powder, building blocks, picking up beans with tweezers, and connecting cables.
- Brain and data remain the key focus globally.** US robotics labs released new models, DYNA-2, Skild AI S1, Generalist GEN-1.5, focused on shortening new-task adaptation, i.e., lowering the marginal cost of teaching a robot a new task. These models reportedly learned from a few examples and composed long-horizon tasks. Previous hardware-focused players are also shifting resources to brain. Unitree plans to allocate 48% of IPO proceeds to robot model R&D. EngineAI unveiled "Awaken," a brain-like five-layer framework. Data remains the bottleneck; we see a shift to egocentric or UMI data. While Chinese players benefited from low labor costs and broader datasets, Figure is pushing general-public data collection with Figure Index. It pays app users to capture first-person/egocentric human-task video to train Helix, aiming to spend US\$1bn over the next 12 months.

MORGAN STANLEY ASIA LIMITED+		
Sheng Zhong		
Equity Analyst		
Sheng.Zhong@morganstanley.com		+852 2239-7821
MORGAN STANLEY & CO. LLC		
Adam Jonas, CFA		
Equity Analyst		
Adam.Jonas@morganstanley.com		+1 212 761-1726
MORGAN STANLEY ASIA LIMITED+		
Carlos Chai		
Research Associate		
Carlos.Chai@morganstanley.com		+852 3963-3180
MORGAN STANLEY & CO. LLC		
William Tackett, CFA		
Research Associate		
William.Tackett@morganstanley.com		+1 212 761-6028
Andrew S Percoco		
Equity Analyst		
Andrew.Percoco@morganstanley.com		+1 212 296-4322
MORGAN STANLEY ASIA LIMITED+		
Chelsea Wang		
Equity Analyst		
Jinlin.Wang@morganstanley.com		+852 2239-1118
Andy Huang		
Research Associate		
Huang.Andy@morganstanley.com		+852 3963-4023
MORGAN STANLEY & CO. LLC		
Daniela M Haigian		
Equity Analyst		
Daniela.Haigian@morganstanley.com		+1 212 761-6071

CHINA INDUSTRIALS	
Asia Pacific	
Industry View	In-Line

In this report you will find:

- Recent Humanoid Updates & Developments
- Global Policy Support Overview
- Humanoid 100 Performance
- Tracking Humanoid Progress
- Global Humanoid TAM/Adoption Estimates

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+ = Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Further comments



Tech Diffusion

A Morgan Stanley Research
Key Theme of 2026

WRC 2026 takeaways – adoption is broadening, productivity matters: Compared to last year, the shift was less about another leap in locomotion and more about whether robots can perform useful tasks, adapt to variation, and approach human productivity. Commercialization in China is broadening, with more customers moving into deployment, but most sites still run tens of robots. Company meetings centered on robot intelligence. VLA, world models, and RL are increasingly combined; the data recipe and deployment feedback loop look more durable than architecture alone. Data remains the bottleneck, yet yield, diversity, and deployment feedback are gaining importance. Plans focus on growth in (1) model parameters, (2) computing power (several players already operate around thousand-card scale), and (3) data collection (targeting 1mn usable hours by end-2026 or 2027).

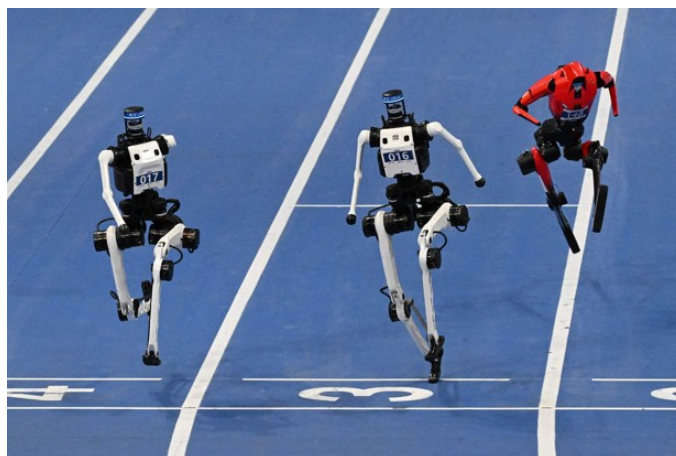
- See our detail note - [WRC 2026 Takeaways](#):

Exhibit 1: World Humanoid Robot Games Leaderboard

Rank	Integrators	Gold	Silver	Bronze	Total
1	-	18	16	12	46
2	Tiangong	15	12	18	45
3	Unitree	3	4	3	10
4	Booster Robotics	3	3	3	9
5	Galbot	2	1	1	4
6	Lumos	1	1	0	2
7	UniX	1	1	1	3
8	Spirit AI	1	1	0	2
9	Noetix	1	0	1	2
10	Galaxea	1	0	0	1
10	Fairino	1	0	0	1
10	Qijiang	1	0	0	1

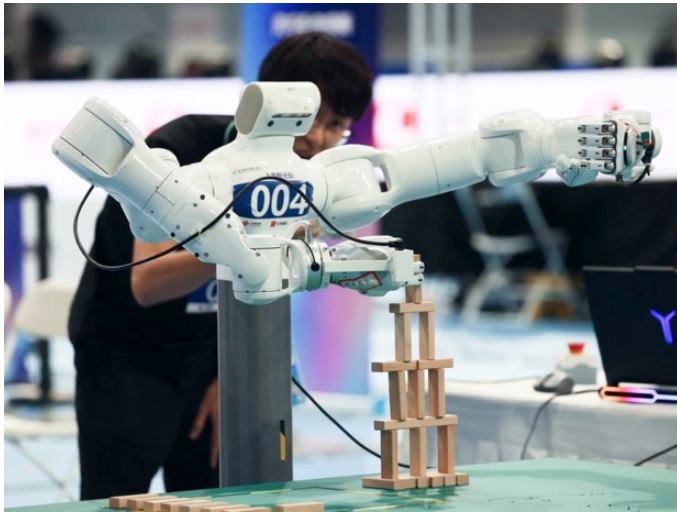
Source: World Humanoid Robot Games, Morgan Stanley Research. Note: Showing top 10 by number of gold medal only.

Exhibit 2: Tiangong ran 100m in 8.64s, breaking Usain Bolt's record of 9.58s



Source: CGTN.

Exhibit 3: Beyond track and field races, this year games also hosted dexterous-hand competitions.



Source: Gasgoo.

Exhibit 4: 2026 Games put much greater weight on multi-step tasks in real-world scenarios vs. single predefined motions.



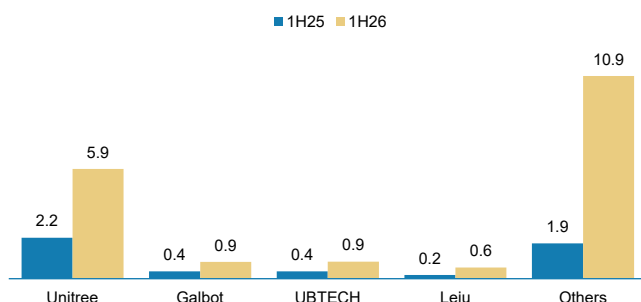
Source: Global Times

1H26 humanoid robots sales volume showed fast growth: Global shipments reached 19k units in 1H26, up 272% YoY, with Chinese players accounting for 97%. Although ~65% of China's shipments remain in entertainment, education/R&D, and data collection, use cases are broadening into industrial and commercial applications. By form factor, half-size bipeds remain the largest in terms of volume (~50%), while wheel-based are expanding rapidly (~29%) in commercial and industrial settings.

- We expect more 1H26 pilots to convert into broader deployments from 2H26, and forecast China's humanoid shipments to reach 50k units in 2026e and 44.6k by 2030e.

Exhibit 5: Per SAG, global humanoid robot shipment reached 19k in 1H26 vs. 5k in 1H25.

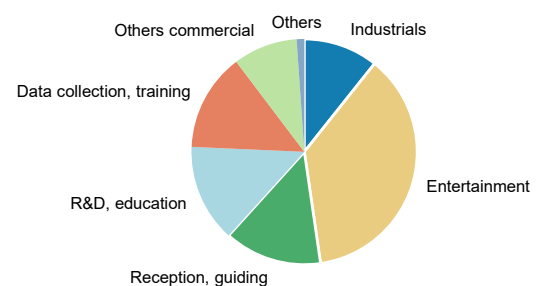
Global Humanoid Shipment ('k unit)



Source: Smart Analytics Global, Morgan Stanley Research.

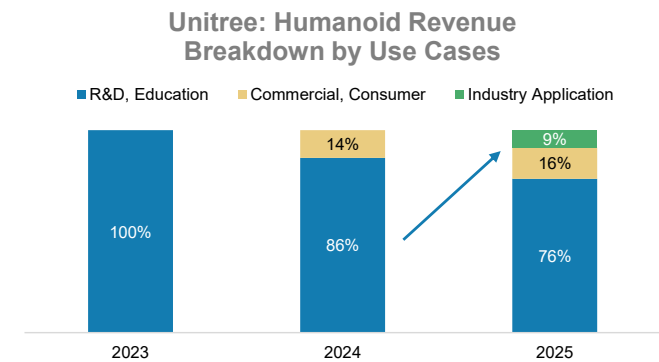
Exhibit 6: We saw more commercial use cases in 1H26 at an industry level...

1H26 China Humanoid Shipment by Applications



Source: MIR Databank, Morgan Stanley Research

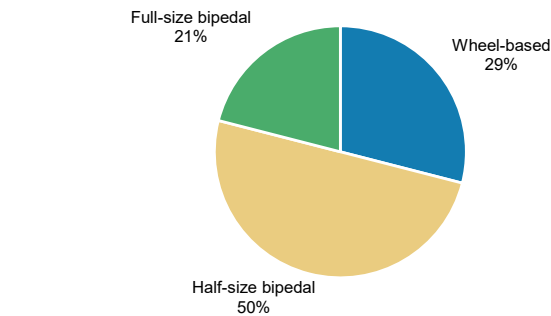
Exhibit 7: ...compared to Unitree's heavy revenue skew toward R&D and education in 2025



Source: Unitree, Morgan Stanley Research.

Exhibit 8: 1H26 – China's humanoid shipments by type

1H26 China Humanoid Shipment by Type

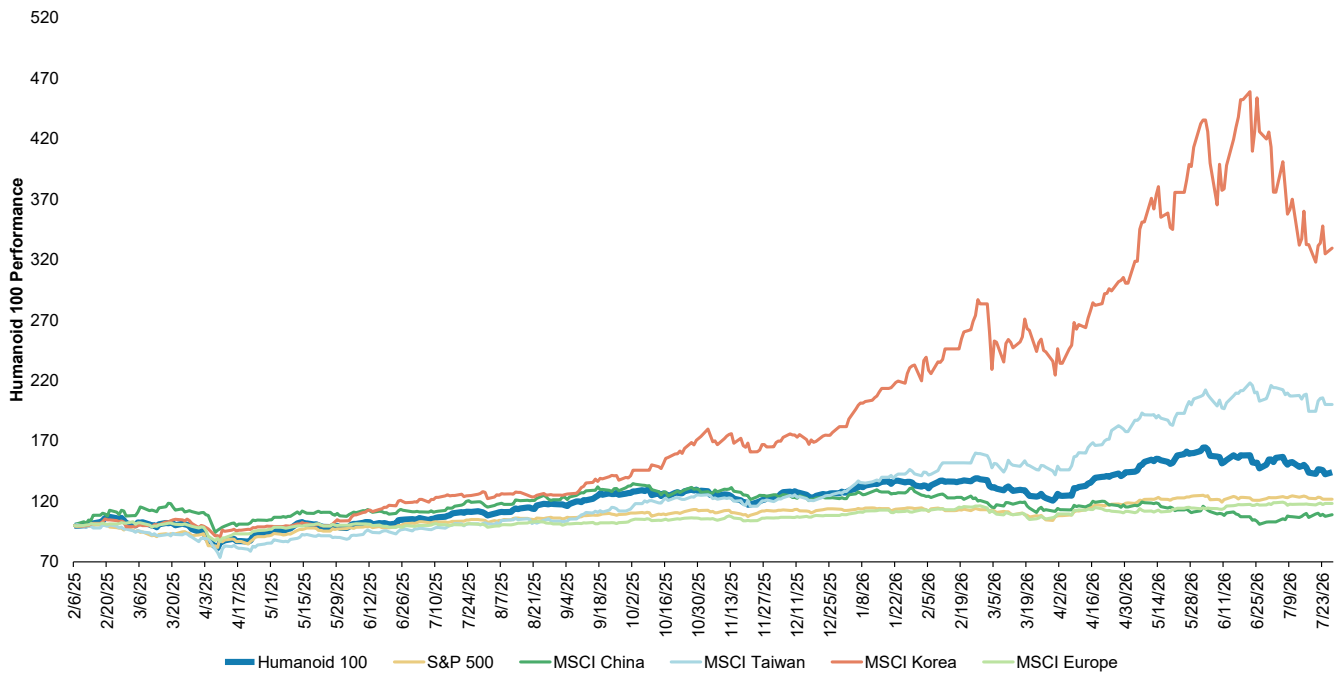


Source: MIR Databank, Morgan Stanley Research

For a copy of our Global Robot Model with forecasts across the full range of embodied AI (not just humanoids), please reach out to your Morgan Stanley representative.

Humanoid 100 update: The equal-weighted Humanoid 100 is up 47% since its inception on February 6, 2025 (adjusting for additions and deletions), outperforming the S&P 500, MSCI Europe, and MSCI China, but underperforming MSCI Korea and MSCI Taiwan. Top performers YTD include Intel, LG Electronics, Arm Holdings, Samsung Electronics, Palo Alto Networks, STMicroelectronics, Teradyne, Flex, Hiwin Technologies, and THK.

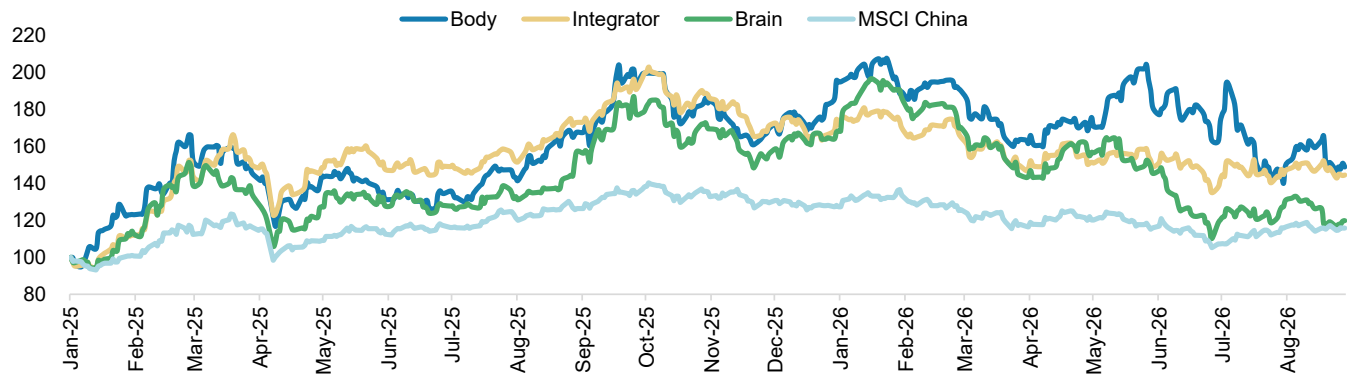
China Humanoid Value Chain update: As of the close on August 28, the China Humanoid Value Chain is down 1.3% MTD on an equal-weighted basis, largely in line with MSCI China -0.5%. But the value chain underperformed YTD at -21.8% vs. MSCI China's -9.1%

Exhibit 9: Equal-Weighted Humanoid 100 Performance since February 6, 2025 (Date List was Established)

Note: Data as of 8/28/2026 close. Price returns only, does not account for dividend reinvestment or transaction costs. Source: FactSet, Morgan Stanley Research.

Exhibit 10: China's Humanoid Value Chain Performance, Equal-weighted

China Humanoid Value Chain - Equal-weighted Performance

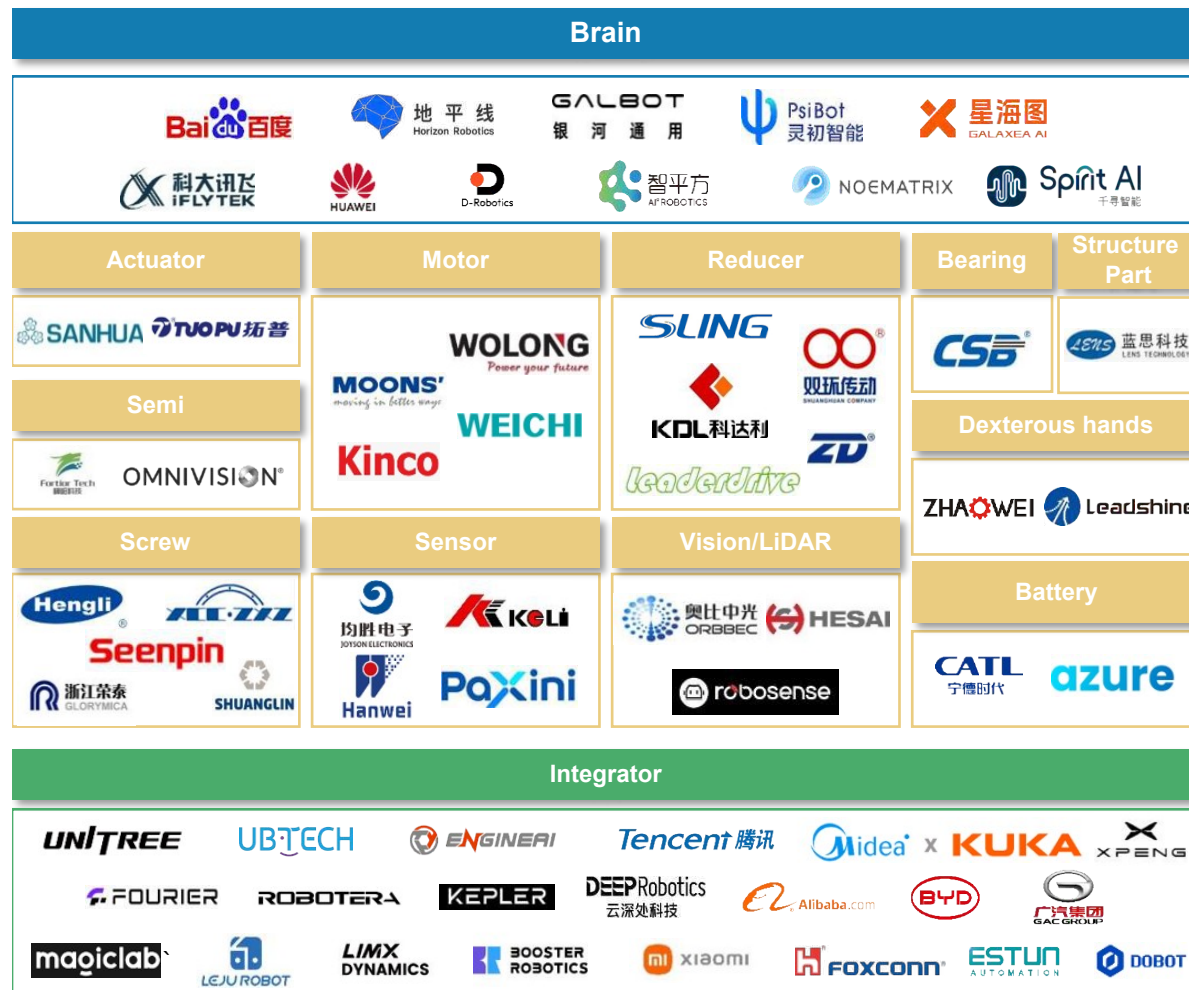


Source: FactSet as of Aug 28, 2026 close, Morgan Stanley Research. Note: the list of companies are before the change of our China Humanoid Value Chain List as stated below. Results shown represent return exclude dividends, brokerage commissions and transaction costs. These figures are not audited. Past performance is no guarantee of future results

Exhibit 11: The Humanoid 100 – Morgan Stanley's List of Global Humanoid Enablers

Brain							Integrators
Foundational Models	Data Science & Analytics	Cybersecurity	Simulation & Vision Software	Semis (Vision & Compute)	Semis (Designers)	Semis (Fab)	OEMs
Baidu Alphabet Microsoft	Palantir ORACLE Microsoft	paloalto Microsoft	HEXAGON Meta Alphabet NVIDIA ROBLOX SIEMENS unity DASSAULT SYSTEMES	intel NVIDIA mobileye Qualcomm Ambarella Horizon Robotics	arm synopsys cadence	TSMC SAMSUNG intel	TESLA BYD Alibaba.com amazon Apple SAMSUNG
Body							
Actuators & Actuator Parts		Sensors	Batteries	Semis (Analog)	Body, Wiring, Thermal	Diversified Automation	
Bearings NSK RKC TIMKEN SKF SCHAEFFLER Regal Rexnord Screws NSK Hengli SKF THK HIWIN Gears / Reducers HIWIN TIMKEN Nabtesco Regal Rexnord		Complete Actuators TIMKEN SANHUA ABB MOOG TUOPU 拓普 Regal Rexnord INOVANCE Motors INOVANCE Regal Rexnord Nidec MOONS Leadshine ZHAOWEI ESTUN RENESAS Encoders Nidec Novanta Sensata Rare-Earths / Magnets Lynas MP MATERIALS JL MAG	Radar & Lidar MAGNA intel TELADYNE TECHNOLOGIES APTIV robosense Magnetic Melexis ALLEGRO Force & Torque Novanta Sensata STE Cameras & Vision Sensors TELADYNE TECHNOLOGIES intel HEXAGON SONY robosense OMNIVISION onsemi KEYENCE	EVE Energy SAMSUNG SAMSUNG SDI LG Energy Solution CATL Semis (Analog) ALLEGRO ANALOG DEVICES infineon RENESAS Melexis OMNIVISION TEXAS INSTRUMENTS ST libra augmented onsemi	Aluminum Castings MAGNA KUSHENG Wires & Connectors Amphenol STE APTIV Thermal TUOPU 拓普 SANHUA	Honeywell Rockwell Automation SIEMENS	RAINBOW ROBOTICS UBTECH GAC GROUP TOYOTA LG HYUNDAI XIBOMI ESTUN AUTOMATION Tencent 腾讯 TERADYNE HEXAGON Alidea NAVER XPENG
							Contract Manufacturers
							FOXCONN PEGATRON LUXSHARE ICT JABIL flex Fosconn Industrial Internet

Note: Public companies only. Not all inclusive. Source: Morgan Stanley Research

Exhibit 12: China Humanoid Value Chain(including public companies in our stock list – see [Exhibit 25](#) and private players)

Source: Morgan Stanley Research

Links of Interest

For Further Reading on Humanoids & Embodied AI

- [WRC 2026 Takeaways: Productivity Matters \(23 Aug 2026\)](#)
- [WAIC 2026 Takeaways: More Physical AI \(20 Jul 2026\)](#)
- [Leader Harmonious Drive Systems: 2Q26 Results Miss on Fast Capacity Expansion \(26 Aug 2026\)](#)
- [China Industrials: Read Across From Unitree IPO \(20 Aug 2026\)](#)
- [China Industrials: Implications from Unitree IPO \(23 Mar 2026\)](#)
- [Humanoids: Meet Unitree G1 Humanoid Robot \(4 Dec 2025\)](#)
- [China Robotics: US Protectionism - Competition Rising \(3 Aug 2026\)](#)
- [Robotics: Thoughts on US Robot Protectionism \(30 Jul 2026\)](#)
- [S. Korea Robotics: US to Restrict Foreign-Made Robots \(29 Jul 2026\)](#)

Sector Blue Paper/Foundation/Insight

- [Rewiring the Home: A Roadmap of Appliances in AI Diffusion \(8 Jul 2026\)](#)
- [Humanoids and Robots – The Next Chapter in China's Dominance in Global Manufacturing \(4 May 2026\)](#)
- [When Factory = Robot \(26 Mar 2026\)](#)
- [Who's Gonna Make the Bots? Adding Contract Manufacturers to the Humanoid 100 \(26 Mar 2026\)](#)
- [World Models: AI's Journey from Digital to Physical \(22 Mar 2026\)](#)
- [Humanoid Tech – Grasping the Future \(30 Nov 2025\)](#)
- [The robots are coming...for critical minerals \(21 May 2025\)](#)
- [Humanoids: A \\$5 Trillion Global Market \(29 Apr 2025\)](#)
- [Investment Implications of Embodied AI \(26 Jun 2024\)](#)

Single Stock Foundation/Insight

- [Tesla Inc: Beyond the Wheel – Mapping Tesla's Journey into Physical AI \(7 Dec 2025\)](#)
- [Apple, Inc.: Apple Robotics — What's Cooking In Cupertino's Kitchen? \(6 Nov 2025\)](#)
- [Hyundai Motor: Hyundai's Robotics Ambition \(17 Aug 2025\)](#)
- [Samsung Electronics: Humanoids – From Digital to Physical AI \(14 Sep 2025\)](#)

Trip/Meetings Takeaways

- [WRC 2026 Takeaways: Productivity Matters \(23 Aug 2026\)](#)
- [WAIC 2026 Takeaways: More Physical AI \(20 Jul 2026\)](#)
- [China Industrials - Humanoids: China Summit Feedback 2026 \(7 Apr 2026\)](#)
- [China Industrials: Humanoid Supply Chain Trip Takeaways: The Emerging Gap \(28 Jan 2026\)](#)
- [The Rowdy Robot: The Robots Take Over Las Vegas \(8 Jan 2026\)](#)
- [China Industrials: Asia Pacific Summit 2025 Feedback – Robotics/Humanoids \(23](#)

Nov 2025)

- China Industrials: Trip Takeaways – Humanoids: Eyes on Use Case Verification (17 Nov 2025)
- China Industrials: Humanoids: How Far Apart Are Reality and Ideal? (12 Nov 2025)
- Humanoids: WRC 2025 Takeaways: Embracing the Future (10 Aug 2025)

Humanoid Horizons

- Humanoid Horizons: Do Robots Have a PR Problem? (28 Jul 2026)
- Humanoid Horizons: Humanoids Coming to Your Bloomberg Screen (May 29)
- Humanoid Horizons: Money Meets Machines (7 May 2026)
- Humanoid Horizons: Can US and China Collaborate on Robots (26 Feb 2026)
- Humanoids: Humanoid Horizons: Gaining Momentum Globally (22 Jan 2026)
- Robotics: Humanoid Horizons: What to Watch for 2026 (19 Dec 2025)
- Humanoids: Humanoid Horizons: Overhyped yet Underappreciated (1 Dec 2025)

The Humanoid Signal

- 2026 Outlook – Humanoids: Commercialization (21 Jan 2026)
- Humanoids: The Tide is Rising, Lifting 2026 Forecast (23 Jun 2026)
- The Humanoid Signal – Broad Verification (5 Feb 2026)
- The Humanoid Signal - How Will the Robot "Brain" Evolve? (26 Feb 2026)
- The Humanoid Signal - Robot Fingers Crossed (6 Mar 2026)
- The Humanoid Signal - Going Global, Capturing Early Ground (14 May 2026)
- The Humanoid Signal: Progress Check, WRC Preview (17 Aug 2026)

The Robot Almanac

- Robotics: The Robot Almanac (Vol. 1): AI Gets Physical; Cambrian Explosion of Bots (7 Dec 2025)
- Robotics: The Robot Almanac (Vol. 2): How to Train Your Robot; Geopolitics; Rare Earths; Sagan's Prophecy (14 Dec 2025)
- Robotics: The Robot Almanac (Vol. 3): Humanoids & Industrial Robots (16 Dec 2025)
- Robotics: The Robot Almanac (Vol. 4): Drones & Air Mobility (17 Dec 2025)
- Robotics: The Robot Almanac (Vol. 5): Space & Defense (18 Dec 2025)
- Robotics: The Robot Almanac (Vol. 6): Autonomous Vehicles (21 Dec 2025)
- Robotics: The Robot Almanac (Vol. 7): Brain Computer Interfaces & Other Form Factors (22 Dec 2025)
- Robotics: The Robot Almanac (Vol. 8): Big Tech's Physical AI Journey (28 Dec 2025)

AlphaWise Survey

- Humanoids: China Humanoids Survey: High Willingness to Adopt Meets Premature Products (2 Dec 2025)

- [Humanoids: China Survey \(2\): What work are humanoids expected to do? \(14 Dec 2025\)](#)
- [Humanoids: Humanoid Robot Consumer and Intern Surveys: China Enthusiasm, US Divided \(4 Aug 2025\)](#)

Relevant Private Company Profiles

- [Robotics: Agile Robots: Das Bot \(17 Mar 2026\)](#)
- [Embodied AI: AI Robotics Disruptors: Skild AI \(23 May 2025\)](#)
- [Embodied AI: Humanoid Disruptors: 1X Technologies \(7 Apr 2025\)](#)
- [Embodied AI: Humanoid Disruptors: Apptronik \(24 Mar 2025\)](#)
- [Embodied AI: Humanoid Disruptors: Unitree - Humanoids Starting at \\$16K \(28 Jan 2025\)](#)
- [Embodied AI: AI Robotics Disruptors: Physical Intelligence \(19 Dec 2024\)](#)

Recent Humanoid Updates and Developments

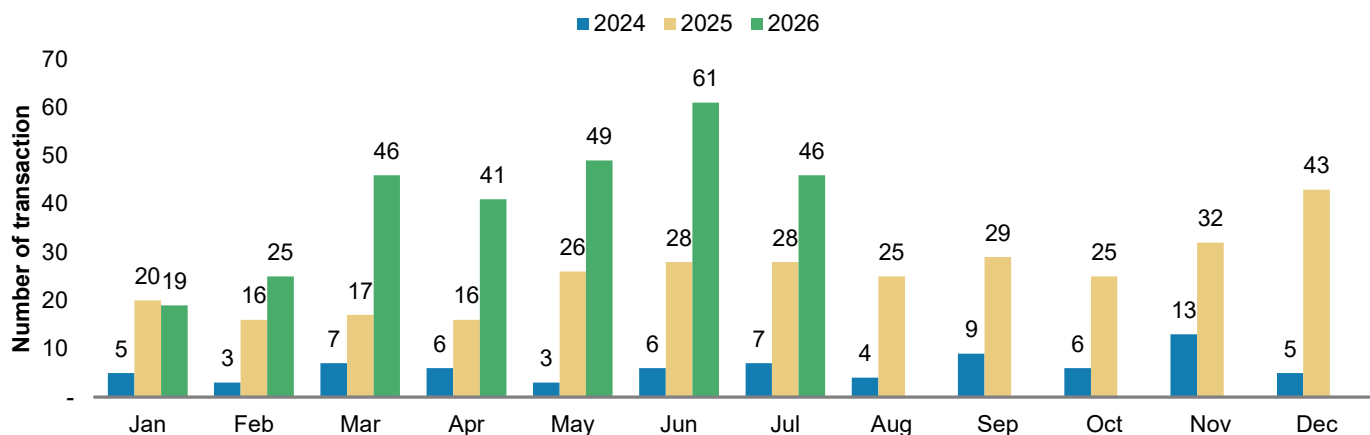
Financing

- Unitree (688836.SS, not covered) completed its IPO in Shanghai's STAR market:** It raised Rmb6bn; the share price surged +460% on the first day of trading; despite falling 30% since then, market cap is now Rmb237bn (US\$35bn). This implies 80x and 48x P/S on FactSet's revenue estimates of Rmb2.9bn and Rmb4.9bn for 2026 and 2027, respectively. Retail enthusiasm was particularly strong, with extremely high public awareness of the Unitree brand, especially with the company appearing on CCTV's Chinese New Year Gala in 2025 and 2026; >8,000x oversubscribed on the retail end. More importantly, we see the industry shifting from education/R&D toward commercial applications, the technology R&D focus pivot toward brains (accounted for 48% of Unitree IPO proceeds) from hardware/locomotion.
 - See our [detailed read-across here](#).
 - With Unitree listed, other Chinese humanoid companies now have a benchmark for valuation; listing activities are likely to continue. LimX Dynamics has reportedly made a confidential Hong Kong IPO filing for up to ~US\$300m, per [Bloomberg](#). Robotera reportedly targets a US\$800m–US\$1bn Hong Kong IPO, per [Bloomberg](#). AI² Robotics [reportedly](#) has begun exploring a 2027 Hong Kong IPO.
- XPENG Robotics raises >US\$900m at >US\$6.3bn post-money valuation.** Outside investors subscribed to US\$600mn of new shares based on US\$6.3bn enterprise value in a round led by IDG Capital with Gaorong Ventures participating and Tencent and Alibaba joining as strategic investors – a marquee cap table rather than strategic or related-party capital. The parent and management subscribed to US\$300mn of new shares on the same terms. Total amount of funds raised reached US\$900mn. This is a positive catalyst for the stock, as the fundraising gives the humanoid robotics business its own currency and external investor base. More importantly, it provides, for the first time, a defensible valuation benchmark to carve out from XPeng's blended valuation.
 - See [detail note by our auto analyst Tim Hsiao](#).
- SoftBank reportedly negotiates majority acquisition of 1X at ~US\$6bn valuation.** SoftBank is reportedly discussing buying control of 1X Technologies. The talks would represent a major consolidation bet on household humanoids if consummated, but discussions are ongoing and neither ownership percentage nor final valuation is fixed.
- Generalist adds nearly US\$200m to Series B at a US\$3bn valuation.** The robot-foundation-model startup reportedly added roughly US\$200m, led by 8VC, after raising US\$400m in June, per [TechCrunch](#). That takes the Series B to roughly US\$600m and lifts Generalist's valuation from US\$2bn to around US\$3bn, reinforcing how aggressively capital is being allocated to hardware-agnostic robot intelligence rather than only robot bodies.
- Mech-Mind to be listed in Hong Kong.** Industrial-robot vision specialist Mech-Mind Robotics (9615.HK) filed its [global-offering documentation](#) for 23.14m H-

shares at HK\$101.7, implying a listing market cap of HK\$12.7bn (before overallotment option is exercised), with listing expected in early September. Mech-Mind recorded revenue of Rmb389mn in 2025 (+47% YoY).

- **Psibot announced a new >US\$100m round** with strategic/industrial investors including Tuopu Group, Chery's Ruicheng fund, Lens Technology, 37 Interactive Entertainment, Wuhu state capital and Fosun. The capital will support its human-data/world-model technology and accelerate commercialization in logistics and advanced manufacturing—particularly relevant because Lingchu is increasingly positioning itself as an intelligence/model layer rather than just another humanoid-body vendor.
 - We met with Mech-Mind and Psibot management in August WRC trip, see [our takeaways here](#).
- **"Humanoid" Raises \$152mn Series A round at reported \$1.35bn valuation.** In late July, UK-based Startup "Humanoid" raised a \$152 million Series A led by Prime Movers Lab with participation from Schaeffler, Bosch, Fubon Financial Holding Venture Capital and Aglaé Ventures, valuing the two-year-old company at \$1.35 billion post-money and taking total funding to \$270 million [according to Reuters](#).

Exhibit 13: Monthly Funding Activity – Chinese Humanoid Companies



Note: Includes all notable funding disclosed based on our knowledge and research. However, some may still not be captured. We have updated the number of transactions in previous months to account for transactions announced after our previously published Humanoids Horizon report. Source: GGII, Company announcements, Morgan Stanley Research

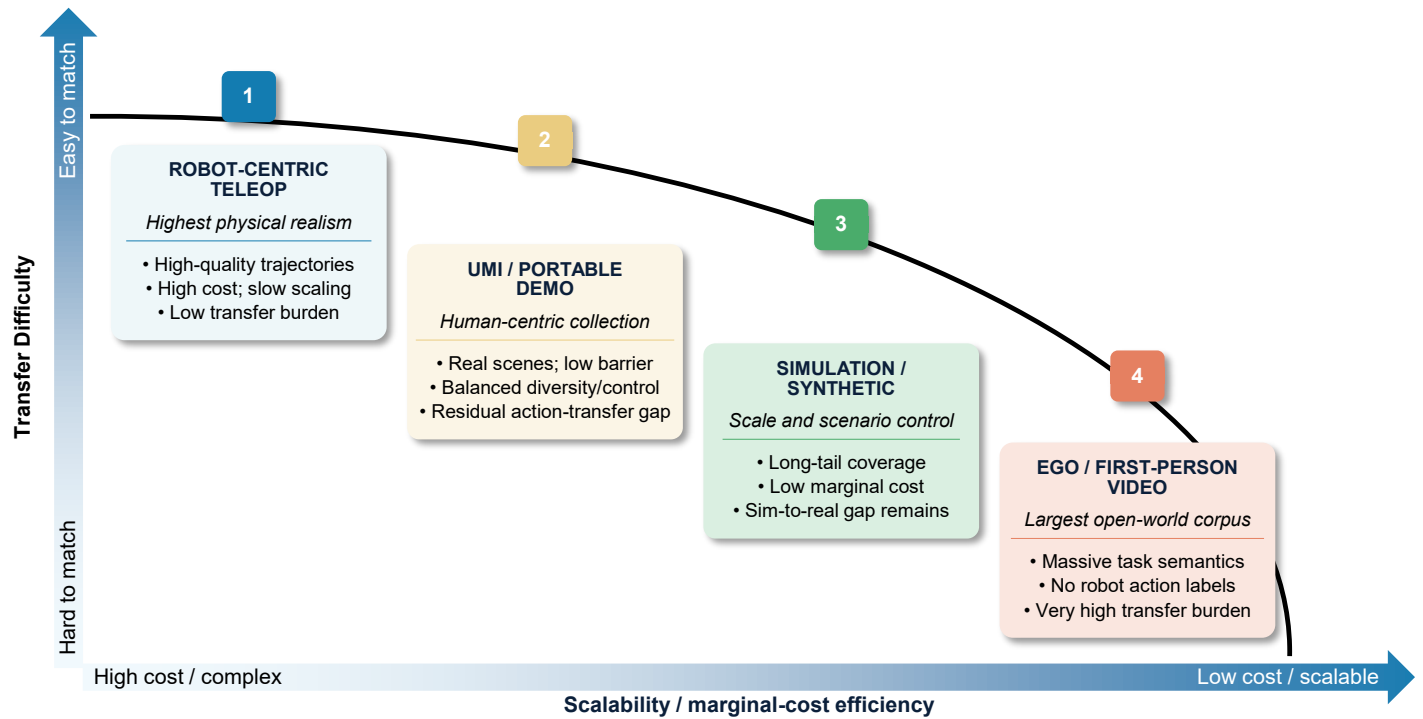
New Robot Models and Hardware Improvements

- **Figure launches "Figure Index," a crowd-sourced physical-world data engine.** [Figure](#) is paying users to capture first-person /egocentric human-task video for training Helix, with the platform already reporting 260k app downloads, with over 44,000 weekly active users and 16mn of videos. Figure paid US\$15mn to creators and plans to spend >US\$1bn over the next 12 months on data and compute, which makes Index one of the clearest attempts to industrialize the humanoid-data flywheel outside China.
- **Hugging Face launches [Microduck](#), a \$399 open-source consumer robot for AI development.** Unveiled August 27 and shipping before Christmas, the 25cm duck can waddle, crouch, pick things up with its beak and right itself after falling, using a camera, LIDAR and two IMUs; behaviors are trained in simulation and deployed

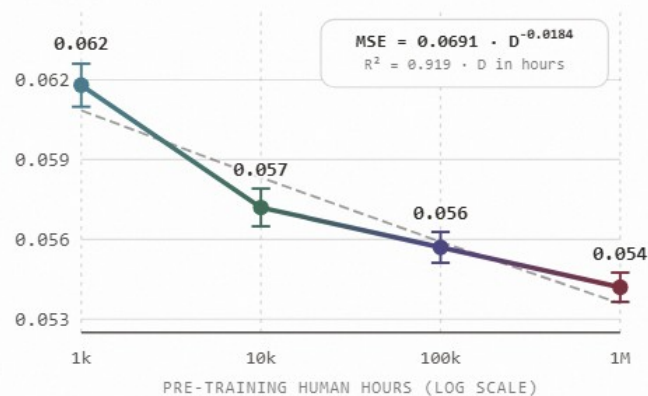
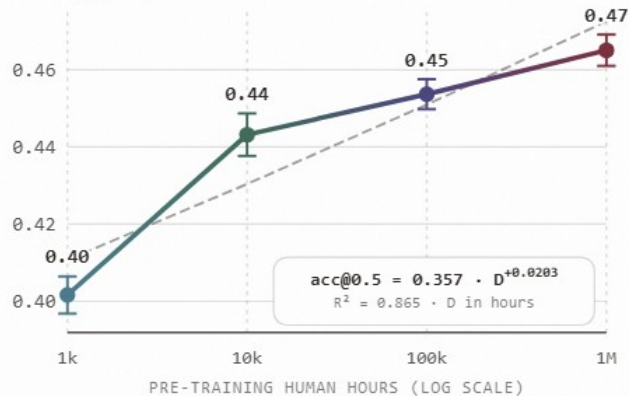
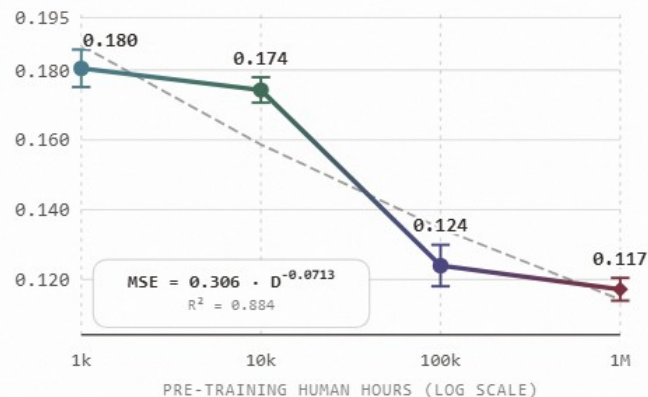
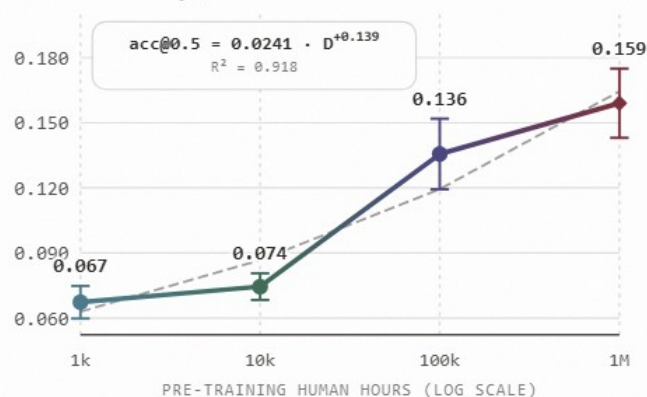
directly to the robot, with the SDK and full RL training stack on GitHub. It follows Hugging Face's 2025 acquisition of Pollen Robotics and its Reachy Mini desktop line, and is notable mainly as evidence of a low-cost, open developer channel forming around physical AI.

- **Dyna releases DYNA-2, trained primarily on >1m hours of human egocentric video.** [DYNA-2](#) is a world-action model (WAM) that uses massive-scale human video pretraining rather than robot demonstrations, followed by relatively limited robot-specific adaptation. Dyna reports a scaling relationship between human-video training and robotic capability and shows transfer across manipulation tasks, i.e. the scaling law for humanoid.
- **Skild AI unveils S1 robotic foundation model.** [S1](#) is designed around in-context learning: Skild demonstrates a robot observing a video/example and attempting unseen tasks without task-specific post-training. S1 can execute previously unseen long-horizon tasks and is already being evaluated with commercial partners.
- **Generalist releases GEN-1.5, emphasizing one/few-shot robot learning.** [Generalist's GEN-1.5](#) can acquire new manipulation skills from extremely short demonstrations, sometimes a single seconds-long example, without per-task fine-tuning, although the tasks are simple and short-horizon. Generalist is betting on a hardware-agnostic "robot brain," rather than vertically integrating its own humanoid.
- **Xiaomi open-sources Robotics-1 training/deployment stack.** The model had already been introduced in July. What changed in August was the open-source release of its real-robot post-training, deployment and evaluation pipeline, built around a model pretrained on >100k hours of human data plus cross-embodiment robot data.
- **EngineAI launches "Awaken" embodied-intelligence engine.** While EngineAI's T800 boxing performance drew most attention in WRC, the company moving toward intelligence stack and industrial tasks such as manipulation, loading and mobility, EngineAI Awaken is a brain-like five layer framework (S1 instinct reflex, S2 action planning, S3 cognitive reasoning, and S4-S5 self-growth with emotional models). Combining WAM and VLA, robot reportedly reached 98% success rate after 2 hours post-training. T800 has been deployed in Luxshare's Suzhou factory, performing material handling and moving task, according to company introduction [[chines source](#)]

Exhibit 14: An overview of the embodied data landscape for training models, mapped across Transfer Difficulty (Y-axis) and Scaling Difficulty (X-axis).



Source: Siyin Wang et. al. - World Action Models: The Next Frontier in Embodied AI, Morgan Stanley Research.

Exhibit 15: Scaling laws for Dyna-2 world-action models trained on a ladder of human video data, from 1k hours to 1mn hours**HELD-OUT HUMAN DATA****Held-out MSE ↓****Accuracy@0.5 ↑****ZERO-SHOT ROBOT DATA (NO ROBOT DATA IN PRE-TRAINING)****Zero-shot action MSE ↓****Zero-shot accuracy@0.5 ↑**

Source: Dyna Robotics - "Dyna-2: A 1-Million-Hour Scaling Law for World-Action Models".

- **Orbbec launches robot-free data collection hardware plus new Physis series vision camera.** The Robot-Free Data Collection Hardware Platform supports high-quality real-world data collection for model training and robot validation, while the Physis series provides wide-field, high-quality visual perception for embodied AI deployment.
- **Galbot debuts ET-1 biped and expands AstraBrain-Agent across robot forms.** Galbot introduced its first bipedal humanoid, ET-1, while emphasizing that the same embodied-intelligence stack can operate across bipedal, wheeled and other forms. AstraBrain-Agent, is an agent architecture designed for real-time multimodal interaction and motion control in humanoid robots. Robot can observe its surroundings continuously, understand spoken instructions, and dynamically plan its actions.
- **DOBOT introduces LUMO, a half-size "all-terrain embodied" humanoid.** LUMO is roughly 1.3m tall and is positioned toward household, education and public-space use rather than heavy industrial manipulation. It combines DOBOT's own embodied-model stack with multimodal perception, navigation/spatial understanding and human interaction, representing another established Chinese

cobot vendor expanding downward into consumer/service humanoids.

Exhibit 16: Orbbec robot-free data collection Hardware Platform (Ego + UMI)



Source: Orbbec

Exhibit 17: Galbot latest ET-1 half-size bipedal humanoid



Source: Galbot

Partnership

- **LG signed an MOU with NVIDIA to build a next-generation biped humanoid** using Isaac GROOT, Jetson Thor and NVIDIA's robotics stack, targeting a public unveiling in Q1 2027. Days later LG detailed a ~10,000sqm robotics "Data Factory" in Seoul that is intended to house hundreds of robots and generate large-scale training data for LG's Robot Foundation Model.

Adoption

- **During Hyundai Motor's Investor Day, HMC stated the target for Atlas deployment at HMGMA from 2028 and US robot production capacity of 30k units annually.** The group also plans to expand RMAC capacity 10x by year-end 2026, while potentially leveraging its manufacturing footprint and dealer/financing networks for commercialization. However, management provided limited additional clarity on the ownership structure of the robotics manufacturing entity and related timeline, which had been a key area of investor interest ahead of the CID.
 - See detailed report by our auto analyst for Korea, Young Suk Shin – [Hyundai Motor - Key Takeaways from CEO Investor Day](#).
- **Schaeffler deepens Hexagon AEON humanoid program ahead of planned ≥1,000-unit rollout.** [Hexagon Robotics](#) and [Schaeffler](#) expanded their partnership, including AEON entering Schaeffler's humanoid gym in Germany. The Humanoid Gym provides a Train–Validate–Deploy model in a dedicated industrial

environment. The companies reiterated plans for at least 1,000 AEON humanoids across manufacturing applications over the coming years.

- **UBTECH and Singtel partner to commercialize enterprise robotics in Singapore/APAC.** The companies signed an [MOU](#) around integrating UBTECH robots with Singtel's enterprise connectivity/services, initially emphasizing enterprise applications and regional deployment. This adds another case for UBTECH overseas expansion.
- **Ant Group's Lingbo robots move into real pharmacy night-shift operations.** [Xinhua](#) reported Lingbot-powered robots operating in Shanghai Guoda pharmacies, autonomously picking medicines during night shifts. The robots use Ant's LingBot embodied-intelligence stack, marking a transition into live commercial operation, not another model demo.

Global Policy Support Overview

Globally, governments are placing greater emphasis on physical AI. To interact with the physical world, humanoid robots rely on a broad array of sensors, including cameras, LiDAR and force/torque sensors, combined with AI models that perceive and map their surroundings. This could raise data security and surveillance concerns, while AI-enabled robots also have clear dual-use characteristics from a national security perspective.

In China, humanoid and embodied intelligence became a clear national priority. Since MIIT issued its Guiding Opinion on Humanoid Innovative Development in February 2023, local governments have released a wide range of policies to support industry development, set up funds for humanoid and embodied-AI ventures, and established institutions and alliances to promote industry cooperation. In 2025, NPC, Li Qiang, the Chinese Premier, first referenced this in the government work report. Subsequently, in the 15th Five-Year Plan announced during the 2026 NPC, robotics became [a strategic emerging industry](#) for the first time, placing it alongside categories including next-gen IT, NEV, biomedicine, and aerospace. Earlier in June, [MIIT and SASAC](#) reportedly announced a new initiative that targets 10k-level deployment capacity and >100 high-value applications by end-2026. Local governments and SOEs must identify operational sites (≥ 20 /province, ≥ 10 /central SOE) for humanoid training.

- **We see continued government leadership and efforts in supporting this frontier industry.** As commercialization begins in 2026, we expect sustained support at both the national and local levels to accelerate development and reinforce China's leadership in this sector, critical to push early adoption and drive the commercialization flywheel.

Exhibit 18: China – national and local policies

National Policies	Release Date	Target
MIIT Guiding Opinions On The Innovative Development Of Humanoid Robots	Jan-23	By 2025, products reach internationally advanced levels and achieve mass production; cultivate 2–3 ecosystem companies with global influence
Standards System For Humanoid Robots And Embodied Intelligence	Feb-26	Covers six pillars: basic commonality, brain/intelligent computing, limbs and components, robots and systems, applications, and safety and ethics
“15th Five-year Plan”	Mar-26	Lists robotics as a strategic emerging industry for the first time
2026 Humanoid Robots And Embodied Intelligence Real-world Training Special Initiative	Jun-26	- By end-2026, humanoid robots complete application validation and routine deployment across a set of representative scenarios - Forming 100+ high-value applications, targeting 10k-level deployment capacity - Identifying real operational sites (≥20 per province, ≥10 per central SOE)
Local Government Policies	Release Date	2027 Target
Zhejiang Province Implementation Plan For Innovative Development Of The Humanoid Robot Industry (2024–2027)	Sep-24	- Provincial annual robot output reaches 20k units - Reaching core-industry scale of Rmb20bn; related-industry scale of Rmb50bn
Shenzhen Embodied Intelligence Robot Technology Innovation And Industrial Development Action Plan (2025–2027)	Mar-25	- Cultivating 10+ new companies valued over Rmb10bn, 20+ companies with revenue over Rmb1bn - Achieving 50+ billion-yuan-level application deployments; related-industry scale of >Rmb100bn
Shanghai Embodied Intelligence Industry Development Implementation Plan	Jul-25	- Reaching industry scale of Rmb50bn - Providing 30% subsidy for model R&D, 20% for application projects, 50% for ecosystem projects
Beijing Economic-technological Development Zone: Measures To Promote Innovative Development Of Embodied Intelligence Robots	Aug-25	- Forming 10k-unit-scale mass-production capability for embodied intelligence robots - Building tens-of-millions-scale real-world embodied dataset
Hangzhou Embodied Intelligence Robot “Chain Strengthening” Action Plan (2026–2027)	Jan-26	- Integrator enterprises’ total output exceeds Rmb20bn - Value-chain total output exceeds Rmb50bn

Source: Government announcement, news report, Morgan Stanley Research

Exhibit 19: Non-China government policies

Country	Date / status	Major policy / initiative	Key details	Targets
South Korea	Apr-25	K-Humanoid Alliance	MOTIE launched a dedicated humanoid alliance with ~40 companies, universities and other organizations.	- Robot-AI foundation model by 2028. - By 2028 humanoid hardware targeted at ≤60 kg weight, ≥50 DoF, ≥20 kg payload, ≥2.5 m/s speed.
	Feb-26	K-Humanoid Moonshot	The government designated humanoids as a core K-Moonshot national mission and established a strategic council. Launched the Public-Private AI Humanoid Core Technology Advancement Project (2026–2030) in May, to invest KRW50.4bn through 2030.	Korea aims to establish humanoid production capacity of 30,000 units annually, by 2030.
Japan	Jun-26	AI Robotics Strategy	Major shift toward Physical AI , domestic multimodal robotics foundation models, robot adoption and large-scale social implementation. Noetra + AIST were selected to lead a domestic multimodal foundation-model project.	Japan aims 10m robots (not only humanoids) deployed by 2040 across 18 sectors, including manufacturing, food, healthcare etc.
US	Jul-26	FCC included foreign-made advanced robotic devices into the Covered List	The addition prohibits FCC authorization, generally blocking importation, marketing, or sale in the US . Robots that had already received FCC authorization are not affected.	-
UK	Jan-26	Robotics Adoption Programme + Regulatory Innovation Office robotics priority	Focus is on accelerating commercial adoption of broad robotics and removing regulatory barriers.	~5 Robotics Adoption Hubs, expected to operate from 2H26. Each receives at least £2m over four years.

Source: Government announcement, news report, Morgan Stanley Research

Humanoid 100 Performance

The Humanoid 100 is Morgan Stanley's list of public companies currently involved in or otherwise materially exposed to the humanoid market. We compiled the list through a mix of discussions with our global team of analysts, conversations with clients on their preferred names to play, and our own proprietary research to find names that are involved in the value chain. We then divided the names into *Brain* (Semis/Software), *Body* (Industrial Components), and *Integrators* (Developing Full Humanoids). For every component of the Humanoid 100, we include details on size/liquidity, core business competencies, rationale for inclusion, and current humanoid involvement if applicable.

For a copy of the underlying Humanoid 100 database or our Global Humanoid TAM model, please reach out to your Morgan Stanley representative.

See initial report: [Humanoids: The Humanoid 100: Mapping the Humanoid Robot Value Chain \(6 Feb 2025\)](#)

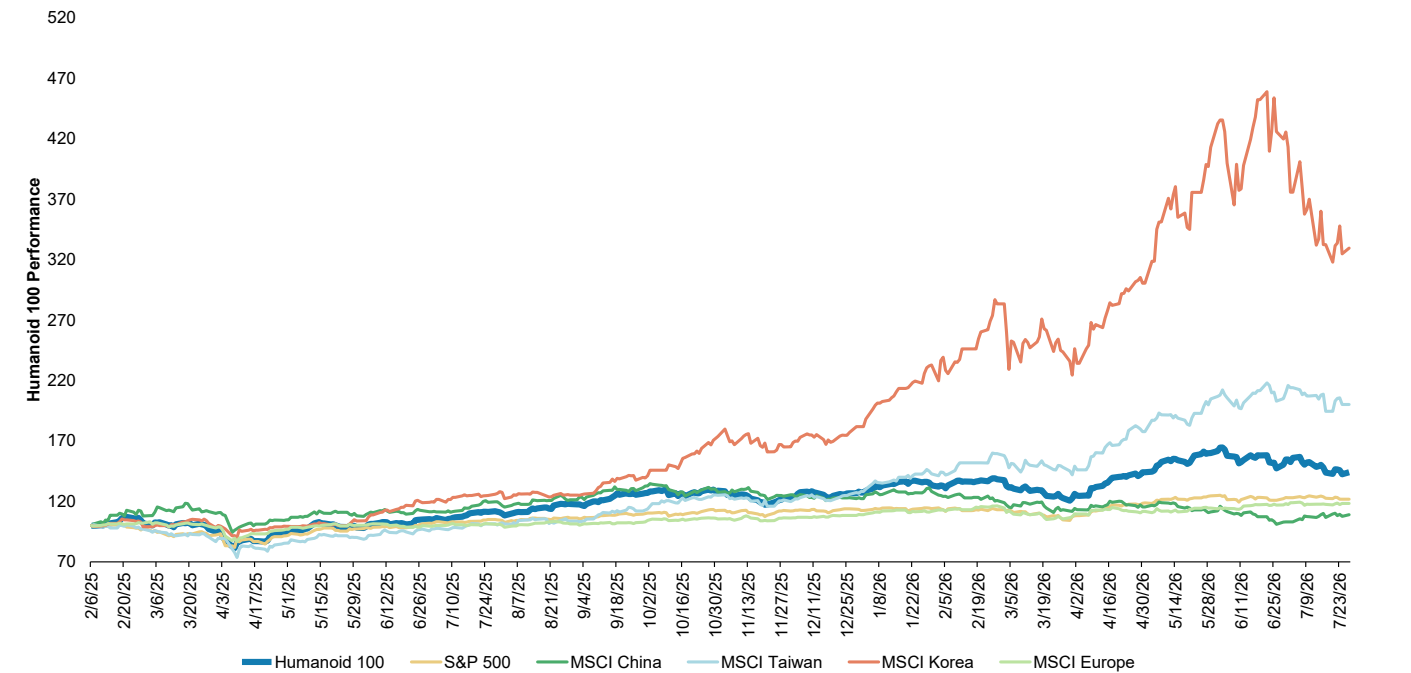
See latest update: [Embodied AI: Who's Gonna Make the Bots? Adding Contract Manufacturers to the Humanoid 100 \(26 Mar 2026\)](#)

Exhibit 20: The Humanoid 100- Morgan Stanley's List of Global Humanoid Enablers

Brain							Integrators
Foundational Models	Data Science & Analytics	Cybersecurity	Simulation & Vision Software	Semis (Vision & Compute)	Semis (Designers)	Semis (Fab)	
百度 Alphabet Microsoft	Palantir ORACLE Microsoft	paloalto Microsoft	HEXAGON Meta Alphabet NVIDIA ROBLOX SIEMENS unity DASSAULT SYSTEMES	intel NVIDIA mobileye Qualcomm Ambarella Horizon Robotics	arm synopsys cadence	TSMC SAMSUNG intel	
Body							
Actuators & Actuator Parts	Sensors	Batteries	Semis (Analog)	Body, Wiring, Thermal	Diversified Automation		
Bearings NSK RBC TIMKEN SKF SCHAEFFLER Regal Rexnord Screws NSK Hengli SKF THK HIWIN Gears / Reducers 绿的谐波 HIWIN TIMKEN Nabtesco Regal Rexnord	Complete Actuators TIMKEN SANHUA ABB MOOG Hengli INOVANCE Regal Rexnord MOONS Motors INOVANCE Regal Rexnord MOONS Leadshine Sensata Technologies ZHAOWEI ESTUN RENASAS Encoders Videx Novanta Sensata Technologies Rare-Earths / Magnets Lynas MP MATERIALS JL MAG	Radar & Lidar MAGNA intel TELETYPE TECHNOLOGIES APTIV robosense Magnetic Melexis ALLEGRO Force & Torque Novanta Sensata E-TE Cameras & Vision Sensors TELETYPE TECHNOLOGIES intel HEXAGON SONY robosense OMNIVISION onsemi E-TE KEYENCE	EVE Energy SAMSUNG SAMSUNG SDI LG Energy Solution CATL ALLEGRO ANALOG DEVICES Infineon RENASAS Melexis OMNIVISION TEXAS INSTRUMENTS ST onsemi	Aluminum Castings MAGNA XUSHENG Wires & Connectors Amphenol E-TE Thermal XUSHENG SANHUA	Honeywell Rockwell Automation SIEMENS	RAINBOW ROBOTICS UBTECH GAC GROUP TOYOTA HYUNDAI XIAOMI ESTUN Tencent TERADYNE HEXAGON Midea NAVER XPENG	
Contract Manufacturers							
FOXCONN PEGATRON LUXSHARE ICT JABIL flex Foscom Industrial Internet							

Note: Public companies only. Not all inclusive. Source: Morgan Stanley Research

Exhibit 21: Equal-Weighted Humanoid 100 Performance since February 6, 2025 (Date List was Established)



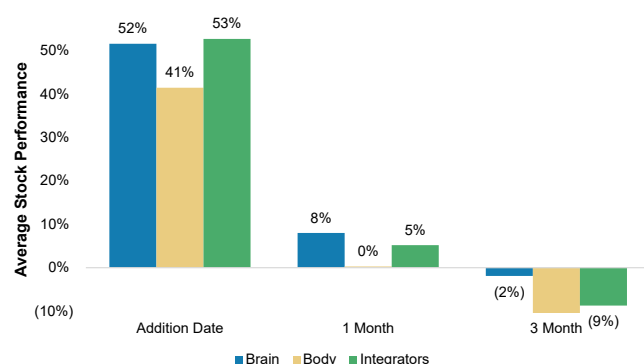
Note: Data as of 8/28/2026 close. Price returns only, does not account for dividend reinvestment or transaction costs. Source: FactSet, Morgan Stanley Research.

Exhibit 22: Top and Bottom Performing Stocks Since Each Stock's Date of Addition to the Humanoid 100 List

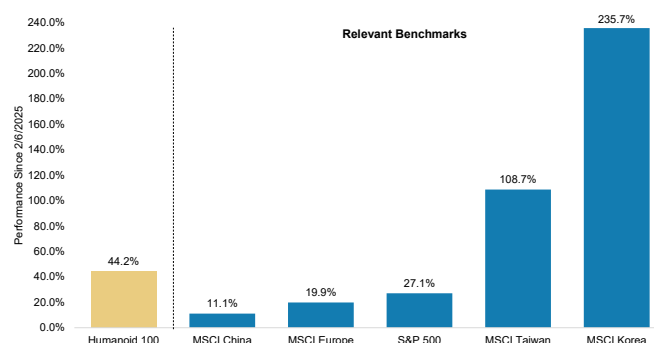
Top Performing Stocks (Since Addition Date)					
Rank	Company	Date Added	Ticker	Country	Performance
1	Samsung Electronics	2/6/2025	005930-KR	Korea	+393%
2	Intel	2/6/2025	INTC-US	USA	+375%
3	Teradyne	2/6/2025	TER-US	USA	+226%
4	Foxconn Industrial Internet (FII)	2/6/2025	601138-CN	Taiwan	+212%
5	Flex	2/6/2025	FLEX-US	USA	+161%
6	Lynas Rare Earths	2/6/2025	LYC-AU	Australia	+151%
7	LG Electronics	2/6/2025	066570-KR	Korea	+146%
8	MP Materials	2/6/2025	MP-US	USA	+145%
9	Amphenol	2/6/2025	APH-US	USA	+130%
10	STMicroelectronics	2/6/2025	STM-US	Switzerland	+126%

Bottom Performing Stocks (Since Addition Date)					
Rank	Company	Date Added	Ticker	Country	Performance
1	Roblox	6/17/2025	RBLX-US	USA	(62%)
2	Robosense	2/6/2025	2498-HK	China	(49%)
3	Mobileye	2/6/2025	MBLY-US	USA	(48%)
4	Dassault Systemes	2/6/2025	DSY-FR	France	(43%)
5	Tuopu	2/6/2025	601689-CN	China	(37%)
6	Moons Electric	2/6/2025	603728-CN	China	(37%)
7	Zhaowei	2/6/2025	003021-CN	China	(36%)
8	GAC Group	2/6/2025	2238-HK	China	(35%)
9	Xusheng	2/6/2025	603305-CN	China	(35%)
10	XPENG	2/6/2025	XPEV-US	China	(34%)

Source: Data as of 8/28/2026 close. Price returns only, does not account for dividend reinvestment. Source: FactSet, Morgan Stanley Research

Exhibit 23: Average Stock Performance across Brain, Body, and Integrator Categories

Note: Data as of 8/28/2026 close. Price returns only, does not account for dividend reinvestment or transaction costs. Source: FactSet, Morgan Stanley Research.

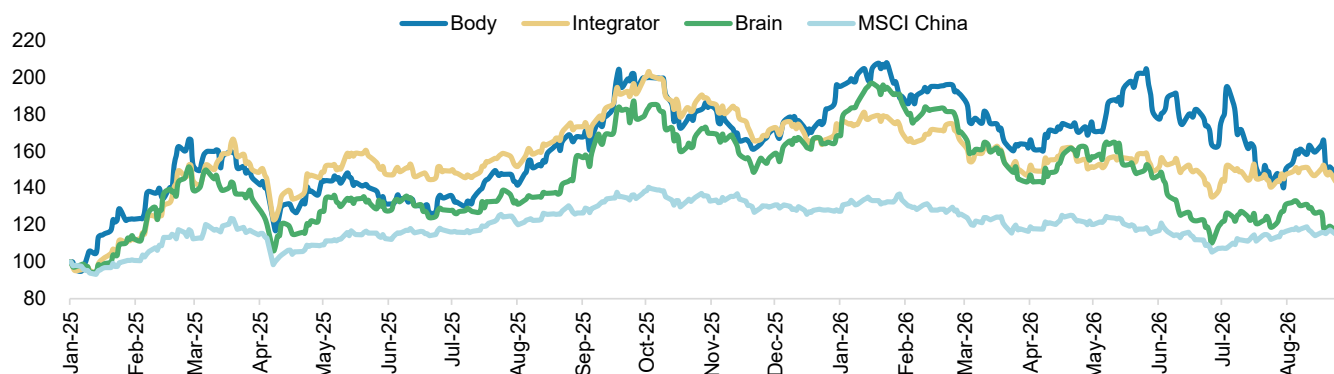
Exhibit 24: The equal-weighted Humanoid 100 is up 44% since February 6, 2025 (adjusting for additions and deletions), outperforming the S&P 500, MSCI Europe, MSCI China, but underperforming MSCI Taiwan and Korea

Note: Data as of 8/28/2026 close. Price returns only, does not account for dividend reinvestment or transaction costs. Source: FactSet, Morgan Stanley Research.

China's Humanoid Value Chain – Performance

Exhibit 25: China's Humanoid Value Chain Performance, Equal-weighted

China Humanoid Value Chain - Equal-weighted Performance

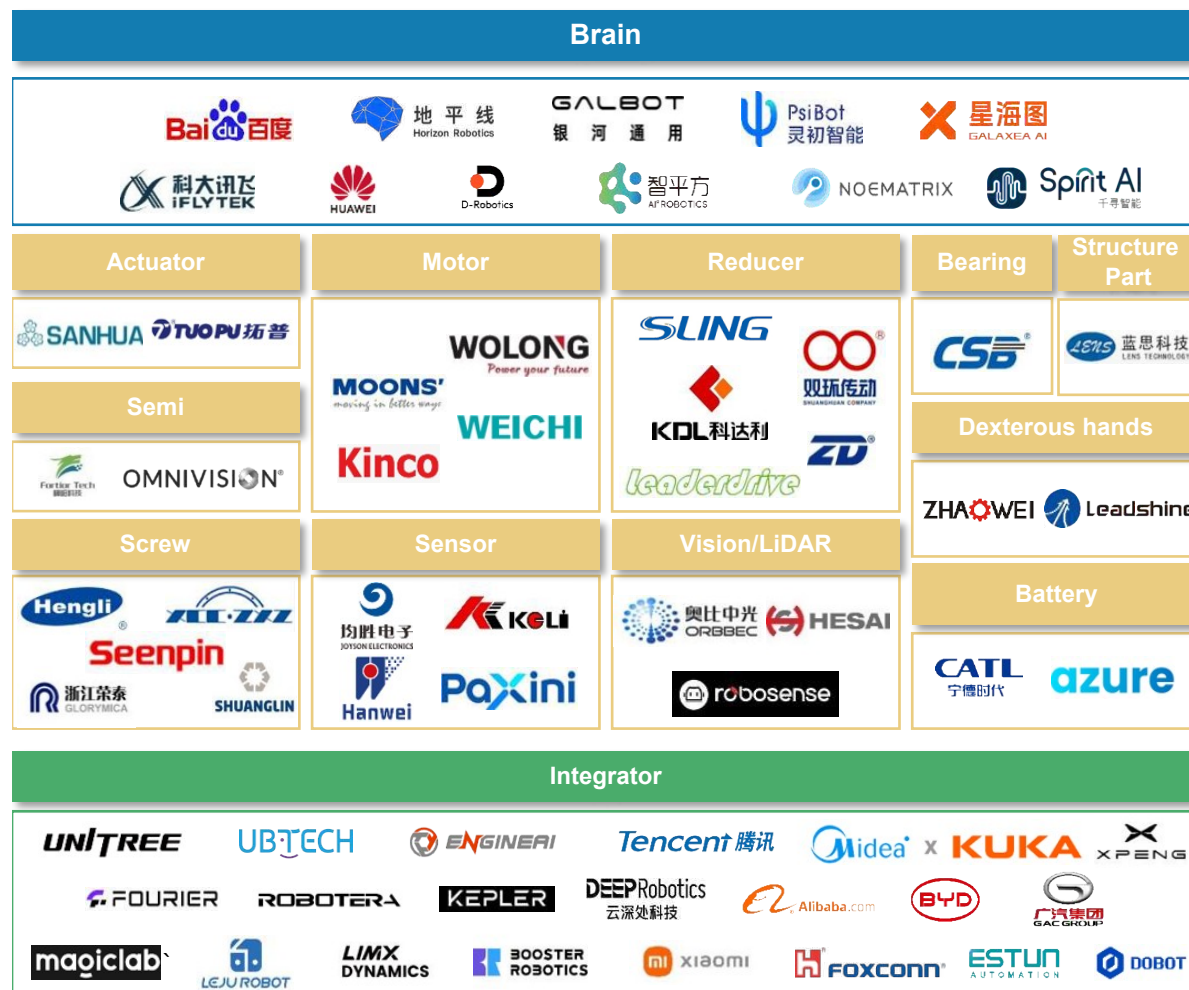


Source: FactSet as of August 28, 2026 close, Morgan Stanley Research. Note: the list of companies is before the change of our China Humanoid Value Chain List as stated below. Results shown represent return exclude dividends, brokerage commissions and transaction costs. These figures are not audited. Past performance is no guarantee of future results

Exhibit 26: China Humanoid Value Chain list: 44 stocks in total, including three in brains, 31 in body components, and 10 integrators

	Ticker	Company	Analyst	Market Cap (USD Bn)	Last Close (LCY)	Price Change				Notes
						1W	3M	YTD	1Y	
Brain	BIDU-US	Baidu	Yu, Gary	26,973	95.3	2%	(28%)	(27%)	(0%)	ERNIE 5.0, a natively multimodal foundation model that supports the "brain/platform" layer.
Brain	002230-CN	iFlytek	Shih, Sharon	13,128	40.3	2%	(15%)	(20%)	(24%)	Deepened cooperation with Beijing Humanoid Innovation Center; and delivered the first ~100-unit soft-core order
Brain	9660-HK	Horizon Robotics	Hsiao, Tim	8,334	4.7	(1%)	(11%)	(45%)	NA	D-Robotics' RDK S100/S600 platforms target on-device robotics workloads including humanoids
Actuator	002050-CN	Sanhua	Wang, Shelley	21,684	36.3	1%	(19%)	(34%)	14%	One of Tesla tier-1 suppliers, developing humanoid actuators
Actuator	601689-CN	Tuopu	Wang, Shelley	12,111	46.8	0%	(25%)	(39%)	(24%)	One of Tesla tier-1 suppliers, developing humanoid actuators
Battery	300750-CN	CATL	Lu, Jack	247,168	363.6	(3%)	(15%)	(1%)	20%	Galbot S1 has reportedly entered CATL's production line, and CATL previously led Galbot financing
Battery	002245-CN	Azure	NC	4,201	17.6	(0%)	(12%)	46%	43%	Azure supplies robot batteries to Unitree, DeepRobotics, Fourier and others
Bearing	300718-CN	Changsheng	NC	3,607	56.4	1%	(23%)	(34%)	(39%)	Provides bearing products, screw products under R&D/sample stage, in collaboration with Unitree
Exterior	300433-CN	Lens	Yang, Derrick	29,796	38.9	3%	(12%)	29%	20%	Supplies domestic customers with joint-module/structural components, and overseas clients with head modules
Hands	003021-CN	Zhaowei	NC	4,504	75.1	3%	(22%)	(40%)	(38%)	Offers dexterous hands, motors and modules; guides revenue to double in 2026, Thailand capacity is largely ready
Hands	002979-CN	Leadshine	NC	2,688	57.2	3%	9%	36%	23%	Frameless torque motor shipments increased to ~120k units in 2025, targeting the million-unit level in 2026
LiDAR	HSAI-US	Hesai	Hsiao, Tim	2,167	16.1	(12%)	(22%)	(28%)	(37%)	1Q26 robotics LiDAR shipments were 118k units, up 137.8% YoY
LiDAR	2498-HK	Robosense	NC	1,116	18.1	(3%)	(44%)	(51%)	(53%)	1Q26 robotics LiDAR shipments were 186k units, up 1458.8% YoY
Magnets	300748-CN	JL Mag	Zhang, Rachel	5,103	27.0	(0%)	(13%)	(21%)	(32%)	Investing US\$144mn in Mexico magnetic-component project, to improve competitiveness in humanoid and EVs
Motor	600580-CN	Wolong	NC	6,833	29.5	(1%)	(24%)	(40%)	(13%)	Received small order for frameless torque motor, partnership with Zhangjiang humanoid center
Motor	603728-CN	Moons	NC	3,185	51.1	9%	(18%)	(29%)	(19%)	Offers coreless motor, frameless torque motor, encoder, reducer products for humanoids
Motor	688698-CN	Veichi	NC	1,383	43.2	(3%)	(33%)	(56%)	(21%)	Formed JV with Rongtai in Thailand for robotics products
Motor	688160-CN	Kinco	NC	1,371	101.0	2%	(5%)	(34%)	12%	Offers frameless torque motor and modules, collaborating with leading humanoid players
Reducer	002472-CN	Shuanghuan	Zhong, Sheng	3,955	35.5	(1%)	(12%)	(25%)	(3%)	Leading supplier of gears, developing new type of reducer for humanoids with a key North America client
Reducer	688017-CN	Leaderride	Zhong, Sheng	8,196	300.5	(0%)	(5%)	56%	99%	Monthly shipment reached c.50k units and capacity expansion continues, involved in module-related projects
Reducer	002850-CN	Kedali	Zhong, Sheng	7,537	176.9	(0%)	3%	12%	28%	Confident in getting orders from leading humanoid integrator. Harmonic reducer has fastest progress.
Reducer	002896-CN	Zhongda Leader	NC	1,897	64.9	0%	(18%)	(27%)	(29%)	Certified supplier for a key domestic integrator, announced to invest Rmb200mn for precision core components
Reducer	301550-CN	Sling	NC	2,730	89.4	4%	(19%)	(7%)	85%	Claims its harmonic reducer will take a significant share in the North American integrator
Screw	601100-CN	Hengli Hydraulic	Zhong, Sheng	20,452	102.5	2%	(9%)	(7%)	12%	Hengli's share in body screw products is indicated at 70%+. The capacity is being prepared mainly in Mexico
Screw	300100-CN	Shuanglin	NC	1,861	21.6	1%	(26%)	(45%)	(54%)	Developed roller screw and grinding machine for humanoid, 10k-unit capacity was planned for Jun-2026
Screw	603119-CN	Rongtai	NC	3,611	51.3	3%	(27%)	(42%)	(15%)	To invest US\$77mn in Thailand for 7mn sets of robotics components, formed JV with Veichi in Thailand for robots
Screw	603667-CN	XCC	NC	2,976	52.2	3%	(26%)	(25%)	33%	To invest Rmb1.1bn for a new plant with an annual capacity of 980k sets of planetary roller screws
Semi	603501-CN	OmniVision	Chan, Charlie	15,471	83.3	(3%)	(17%)	(34%)	(43%)	Its existing sensor technology can be easily applied to humanoid's vision
Semi	688279-CN	Fortior	NC	3,090	197.5	8%	(6%)	(3%)	(4%)	Offers motor ASIC, driver IC, MCU, and established JV with Sanhua to develop dexterous hands
Sensor	600699-CN	Joyson	Wang, Shelley	4,103	19.2	0%	(29%)	(39%)	(9%)	Developed 6-axis force sensors, tactile sensors, IMU, and has placed commercial orders for humanoid robots
Sensor	603662-CN	Keli	NC	2,190	52.4	7%	(16%)	(27%)	(27%)	Developed 6-axis force sensor, and deploying humanoid robots in production line
Sensor	300007-CN	Hanwei	NC	1,549	31.8	2%	(23%)	(41%)	(32%)	Focus on soft, flexible sensor, cooperating with multiple leading integrators
Vision	688322-CN	Orbbec	NC	3,900	101.5	5%	(10%)	13%	18%	A key structured light camera supplier to leading domestic players
Vision	688400-CN	Luster LightTech	NC	3,300	48.5	5%	(12%)	14%	25%	Providing motion capturing solution - FZMotion for Unitree
Integrator	700-HK	Tencent	Yu, Gary	526,032	453.0	2%	(3%)	(24%)	(25%)	Tencent established a Robotics X lab in Shenzhen in 2018, unveiled XiaoWu
Integrator	BABA-US	Alibaba	Yu, Gary	289,677	114.0	(5%)	(10%)	(22%)	(16%)	Released Qwen Studio, a foundation model for physical intelligence, Ant Group's Robbyant unveiled LingBot VA
Integrator	1810-HK	Xiaomi	Meng, Andy	75,151	27.6	(1%)	(3%)	(30%)	(49%)	Unveiled Xiaomi-Robotics-0, an open-source VLA model
Integrator	1211-HK	BYD	Hsiao, Tim	86,743	87.2	(7%)	(6%)	(9%)	(20%)	Developing humanoid project - Yao Shun Yu, testing UBTECH humanoids; invested in Paxini
Integrator	300-HK	Midea/KUKA	Lou, Lillian	98,102	101.4	3%	11%	19%	N/A	The self-developed 6-arm wheel-based humanoid Robot Ultra has entered its washing machine factory
Integrator	XPEV-US	XPENG	Hsiao, Tim	8,978	11.4	(2%)	(35%)	(44%)	(46%)	Its humanoid robot - Iron, is expected to enter mass production stage by end-of-2026
Integrator	002747-CN	Estun	Zhong, Sheng	3,898	31.7	(0%)	9%	34%	34%	Codroid 02 was unveiled in Jun-2025 for flexible manufacturing
Integrator	9880-HK	UBTECH	NC	4,740	84.8	4%	(24%)	(33%)	(15%)	Walker S2 has begun mass production and delivery, targeting 5,000 annual industrial humanoids by 2026
Integrator	2432-HK	Dobot	NC	1,195	23.8	(4%)	(24%)	(37%)	N/A	Dobot started mass production and delivery of the third batch of industrial humanoid ATOM robots in Feb-2026
ODM	002600-CN	Lingyi iTech	Shih, Sharon	14,381	12.8	5%	(14%)	(18%)	(18%)	ODM for multiple integrators, first humanoid robots rolled off Lingyi's Beijing Super Factory

Source: FactSet as of August 31, 2026 close, Morgan Stanley Research. Note: This is a constantly evolving list of Chinese companies involved in humanoids. Companies are included based on the following criteria: (1) MS covered companies; (2) Non-MS covered companies - either announced cooperation with key humanoid companies, or a certain revenue level in products to be applied in humanoid robots. Results shown represent return excluding dividends, brokerage commissions and transaction costs. These figures are not audited. Past performance is no guarantee of future results. For important disclosures regarding companies that are the subject of this screen, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/researchdisclosures. ++ Rating and estimates for this company have been removed from consideration in this report because, under applicable law and/or Morgan Stanley policy, Morgan Stanley may be precluded from issuing such information with respect to this company at this time.

Exhibit 27: China Humanoid Value Chain(including public companies in our stock list - see [Exhibit 25](#) and private players)

Source: Morgan Stanley Research

Exhibit 28: Robot makers comps.

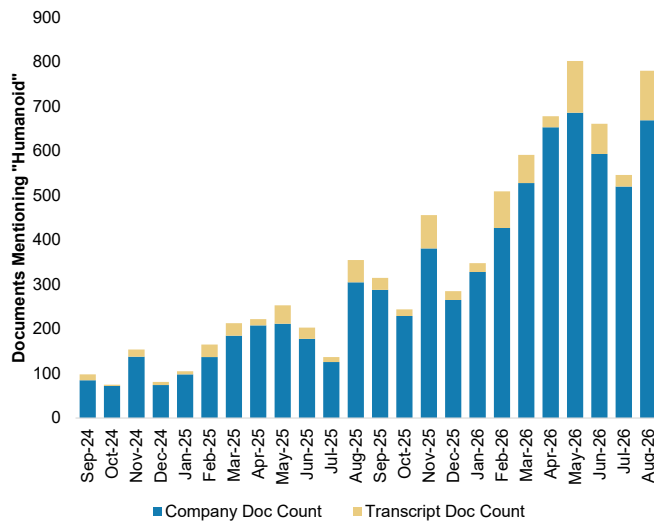
Ticker	Company Name	MS Rating	Trading Curr	Last close 8/28/26	Mkt Cap US\$ (mn)	EPS Growth (%)		P/E (x)		P/B (x)		P/S (x)		EV/EBITDA (x)		ROE (%)	
						2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E
China Industrial Robot / Cobot																	
002747.SZ	Estun	EW	CNY	31.5	3,985	474%	59%	106x	67x	13x	12x	5x	4x	66x	53x	12%	16%
2432.HK	Dobot	NC	HKD	23.8	1,195	-	-	-	-	4x	4x	10x	7x	-	-	-5%	-2%
1021.HK	Huayan	NC	HKD	9.1	624	-	230%	192x	58x	4x	4x	7x	5x	173x	39x	2%	5%
Average						474%	145%	149x	62x	7x	7x	7x	5x	119x	46x	3%	6%
Specialized Robot																	
2590.HK	Geekplus	OW	HKD	10.1	1,740	-	-	-	44x	3x	3x	3x	2x	69x	28x	-1%	9%
6600.HK	OneRobotics	NC	HKD	53.3	1,532	-	-	-	91x	7x	7x	7x	4x	8,138x	195x	0%	6%
2729.HK	Galaxis	NC	HKD	33.0	1,799	-	-	-	-	-	-	10x	8x	-	-	-	-
Average						-	-	-	67x	5x	5x	7x	5x	4,104x	111x	0%	7%
Humanoid Robot																	
9880.HK	UBTECH	NC	HKD	83.2	4,651	-	-	-	160x	6x	6x	7x	5x	-	51x	-4%	3%
688836.SS	Unitree	NC	CNY	585.3	35,205	120%	87%	374x	200x	70x	52x	80x	48x	305x	168x	19%	26%
Average						-	-	-	180x	38x	29x	44x	26x	-	110x	7%	14%
Overseas Industrial Robot																	
6954.T	Fanuc	EW	JPY	6052.0	35,471	21%	14%	28x	25x	3x	3x	6x	5x	20x	18x	11%	12%
6506.T	Yaskawa	OW	JPY	4811.0	8,097	44%	13%	25x	22x	2x	2x	2x	2x	14x	13x	10%	11%
ABB.N.S	ABB	EW	CHF	79.3	178,422	24%	24%	25x	20x	10x	9x	5x	4x	21x	17x	34%	39%
Average						30%	17%	26x	22x	5x	5x	4x	4x	18x	16x	18%	20%

Source: Factset as of Aug 28 close, Morgan Stanley Research. MSe for covered companies, Factset consensus for non-covered (NC) companies.

Tracking Humanoid Progress

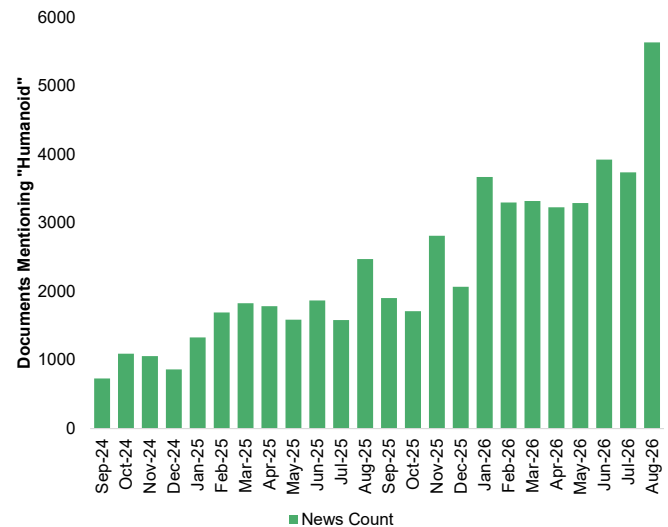
Humanoid Company & News Mentions

Exhibit 29: Time series of company transcripts mentioning "Humanoid"



Note: 'Company Documents' include filings, presentations, and press releases. Data is global. Source: AlphaSense, Morgan Stanley Research

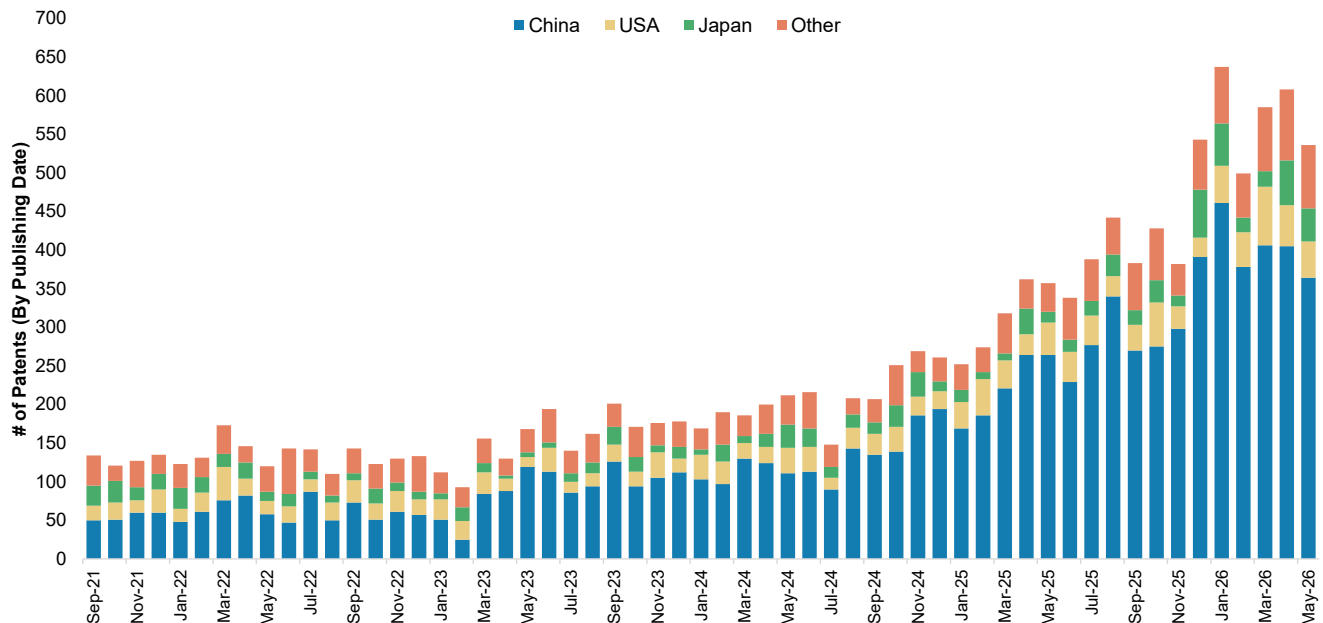
Exhibit 30: Time series of news/media reports mentioning "Humanoid" (increase appears related to WRC)



Note: Data is global. Source: AlphaSense, Morgan Stanley Research

Global Patents

Exhibit 31: Monthly Patent Applications Mentioning "Humanoid" by Country



Note: Ordered by publishing date. According to Google Patents, may not capture 100% of patents. Source: Google Patents, Morgan Stanley Research

Exhibit 32: Global Patent Applications Mentioning "Humanoid" by Office Over the Past 5 Years (According to Google Patents, may not capture all)

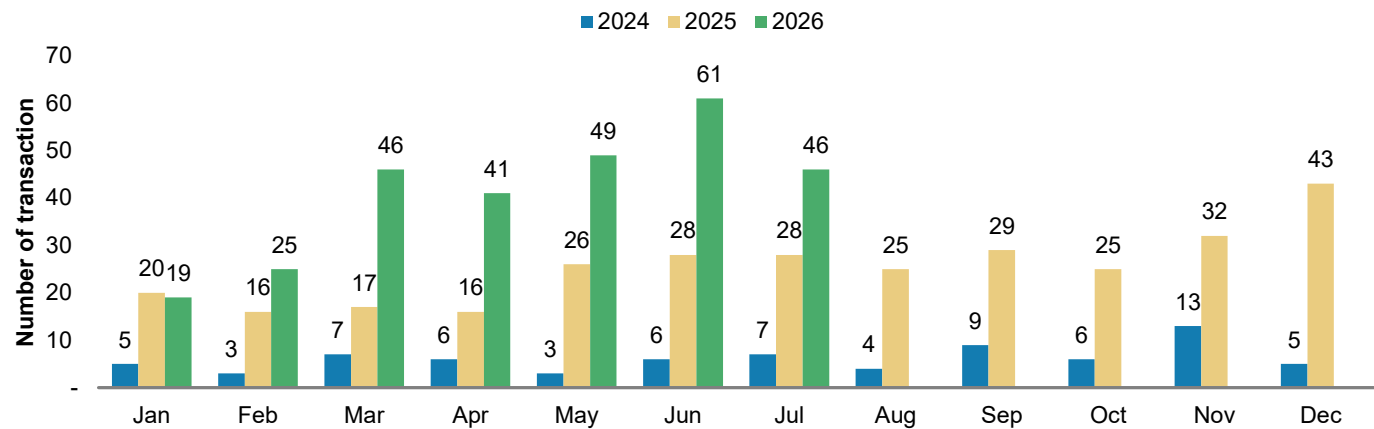
Patents Mentioning "Humanoid" By Office (Past 5 Years)

Rank	Patent Office	# Count
1	China	8,838
2	United States	1,646
3	World Intellectual Property Organization (WIPO)	1,117
4	Japan	1,115
5	South Korea	339
6	European Patent Office (EPO)	228
7	Taiwan	206
8	Germany	126
9	Australia	27
10	Brazil	24
11	Canada	16
12	Poland	16
13	Turkey	15
14	Great Britain	13
15	France	11
16	Italy	10
17	Spain	6
18	Eurasian Patent Organisation (EAPO)	6
19	Sweden	4
20	Switzerland	3

Note: Exhibit limited to top 20. Source: Google Patents, Morgan Stanley Research

China – Financing Activity

Exhibit 33: Monthly Funding Activity – Chinese Humanoid Companies



Note: Includes all notable funding disclosed based on our knowledge and research. However, some may still not be captured. We have updated the number of transactions in previous months to account for transactions announced after our previously published Humanoids Horizon report. Source: GGII, Company announcements, Morgan Stanley Research

Tech Firms Supporting the Development of Humanoids

Exhibit 34: Humanoid or Robotics Involvement of Major Tech Firms in the US & China**Humanoid or Robotics Involvement of Big Tech Firms**

Company	Involvement
 	In Sept, ANT Group's robotics unit, Robbyant, unveiled 'R1' robot at trade show in Shanghai
	Major in-house robotics team and deep experience with automation. Backer and partner of Agility Robotics. FAR (Frontier AI Robotics) Team building Robot Foundation models.
	Reportedly planning a tabletop robot and have explored humanoid designs. Have released numerous AI frameworks & datasets for humanoid and non-humanoid robots.
	UBTech's humanoids leverage Baidu's Ernie AI model. DEEPX leverages Baidu PaddlePaddle deep learning framework.
	Unveiled general purpose VLA model ('GR3') and ByteMini robot
	Google Deepmind partnered with Apptронik and Others. Gemini Robotics is a major focus within Google Deepmind. Also develop MuJoCo physics engine for robotics.
	Embodied AI Innovation Center' in Shenzhen. Develop CloudRobo embodied intelligence platform based on Pangu AI model. Partnered with UBTech.
	Reportedly forming new team within Reality Labs to conduct robotics work. CTO Andrew Bosworth described humanoids as 'next AR-sized bet'.
	Partner and backer of Figure AI. Have an internal autonomous systems & robotics research team.
	Partnered with majority of humanoid developers under Project Gr00t. Bulk of humanoid players leverage Isaac Lab, Gr00t Models, Cosmos, or other tools.
	In-house robotics team working on humanoid and non-humanoid robots. Tencent Cloud has partnership with Dobot.
	Developing 'Optimus' humanoid.
	Developing 'CyberOne' humanoid

Source: Company announcements, Bloomberg, Reuters, Morgan Stanley Research

Global Humanoid TAM/Adoption Estimates

Note: The underlying Excel file for the TAM model discussed in this section is available upon request. Please reach out to your Morgan Stanley representative to obtain the model.

Global Adoption Estimates

According to Our Estimates:

- **By 2036, ~28.1 million (~2.8% of ~1 billion total) humanoid adoptions will have occurred.**
 - Low Income Countries: ~0.2 million (~1.8% of ~14 million total) humanoids have been adopted.
 - LI Countries Include: Rwanda, Ethiopia, Madagascar, Afghanistan, Yemen, etc.
 - Lower Middle Income Countries: ~3.7 million (~1.8% of ~204 million total) humanoids have been adopted.
 - LMI Countries Include: India, Honduras, Vietnam, Zimbabwe, Egypt, etc.
 - Upper Middle Income Countries (Inc. China): ~16.7 million (~3.3% of ~504 million total) humanoids have been adopted.
 - UMI Countries Include: China, Brazil, Belarus, Indonesia, South Africa, Mexico, etc.
 - High Income Countries (Inc. US): ~7.4 million (~2.5% of ~296 million total) humanoids have been adopted.
 - HI Countries Include: USA, Australia, Germany, Canada, United Arab Emirates, Japan, etc.
- **By 2040, ~138.5 million (~13.6% of ~1 billion total) humanoid adoptions will have occurred.**
 - Low Income Countries: ~1.1 million (~7.8% of ~14 million total) humanoids have been adopted.
 - Lower Middle Income Countries: ~16.2 million (~7.9% of ~204 million total) humanoids have been adopted.
 - Upper Middle Income Countries (Inc. China): ~81.4 million (~16.1% of ~504 million total) humanoids have been adopted.
 - High Income Countries (Inc. US): ~39.7 million (~13% of ~296 million total) humanoids have been adopted.
- **By 2044, ~430 million (~42.2% of ~1 billion total) humanoid adoptions will have occurred.**
 - Low Income Countries: ~3.9 million (~27.9% of ~14 million total) humanoids have been adopted.
 - Lower Middle Income Countries: ~57.6 million (~28.1% of ~204 million total) humanoids have been adopted.
 - Upper Middle Income Countries (Inc. China): ~245.7 million (~48.7% of ~504 million total) humanoids have been adopted.
 - High Income Countries (Inc. US): ~123.2 million (~41.6% of ~296 million total) humanoids have been adopted.
- **By 2050, ~1 billion humanoid adoptions will have occurred.**

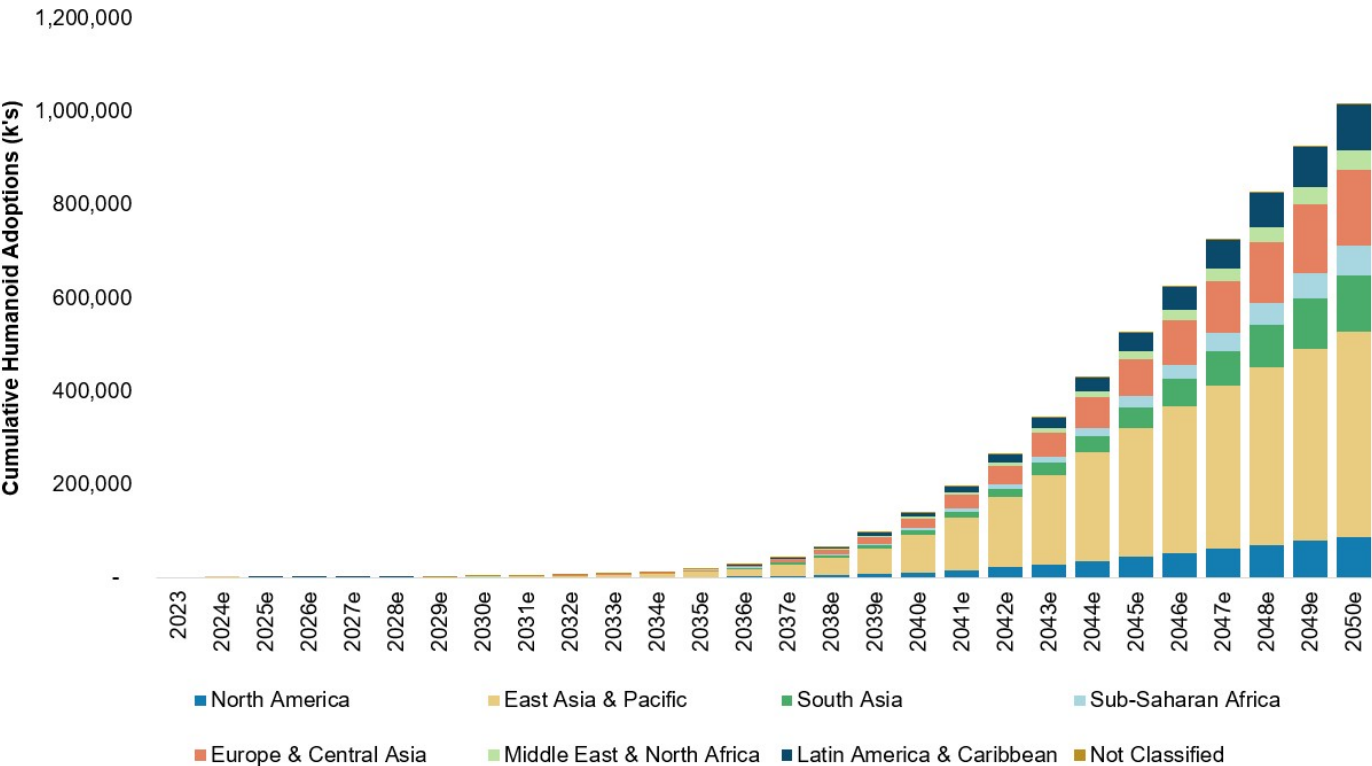
- ° Low Income Countries: ~14 million humanoids have been adopted.
- ° Lower Middle Income Countries: ~204 million humanoids have been adopted.
- ° Upper Middle Income Countries (Inc. China): ~504 million humanoids have been adopted.
- ° High Income Countries (Inc. US): ~296 million humanoids have been adopted.

Exhibit 35: Key Adoption Estimates

Cumulative Adoptions (k's)				
Classification	2036	2040	2044	2050
Low Income	246	1,099	3,901	14,006
Lower Middle Income	3,745	16,248	57,553	204,498
Upper Middle Income	16,739	81,411	245,743	504,307
High Income	7,363	39,714	123,249	296,427
North America	2,159	11,512	35,816	86,866
East Asia & Pacific	15,872	78,971	234,002	439,936
South Asia	2,258	9,797	34,705	122,733
Sub-Saharan Africa	1,137	4,967	17,617	62,938
Europe & Central Asia	3,902	20,617	64,798	163,370
Middle East & North Africa	861	4,146	13,871	42,532
Latin America & Caribbean	1,862	8,289	29,032	98,361
Not Classified	41	173	607	2,501
USA	1,934	10,289	32,017	77,703
China	13,002	65,131	187,937	302,316
Other	13,157	63,051	210,492	639,219
Commercial	26,475	131,799	409,823	935,052
Household	1,618	6,673	20,623	84,185
Global	28,093	138,471	430,447	1,019,238

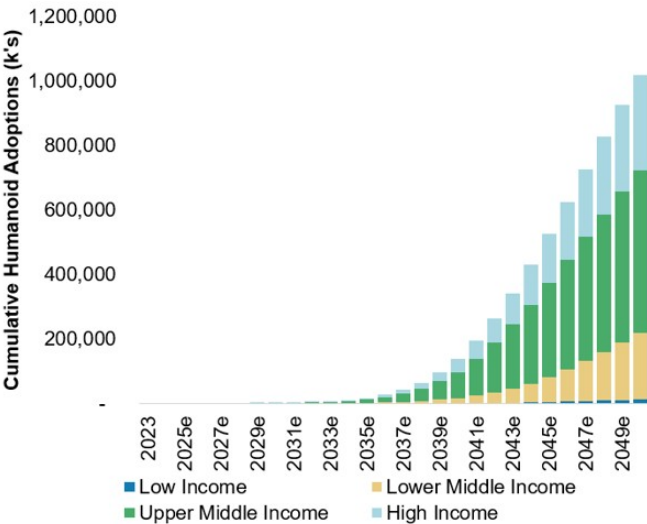
Source: Morgan Stanley Research estimates

Exhibit 36: Cumulative Humanoid Adoptions by Region



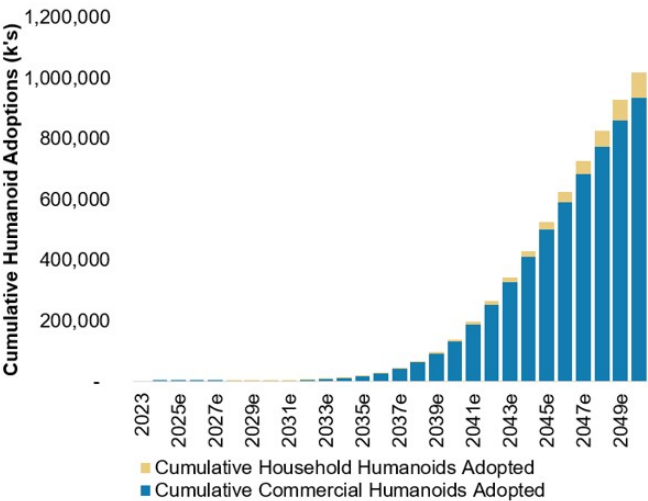
Source: Morgan Stanley Research estimates

Exhibit 37: Cumulative Humanoid Adoptions by Country Income Classification (United Nations)



Source: Morgan Stanley Research estimates

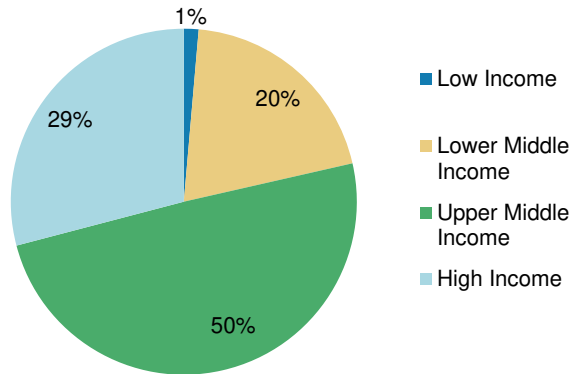
Exhibit 38: Cumulative Humanoid Adoptions: Household vs. Commercial End-Markets



Source: Morgan Stanley Research estimates

Exhibit 39: 2050 Cumulative Adoptions by Income Classifications

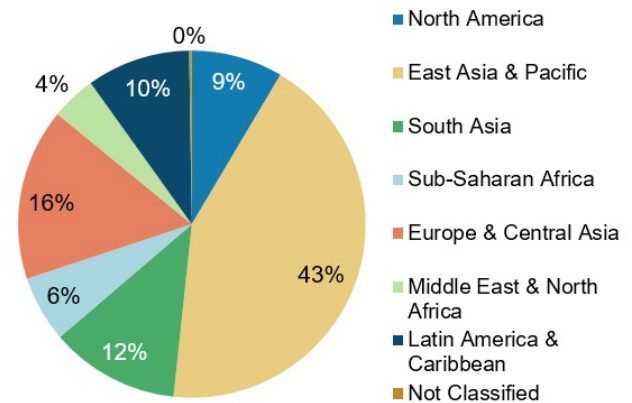
% of 2050 Cumulative Adoptions



Source: Morgan Stanley Research estimates

Exhibit 40: 2050 Cumulative Adoptions by Region

% of 2050 Cumulative Adoptions



Source: Morgan Stanley Research estimates

Sizing the Global Market

We estimate that the global humanoid market could reach \$7.5 trillion in annual revenue by 2050. Assuming a six-year replacement cycle and the below ASP assumptions, we estimate that the global humanoid market could reach \$7.5 trillion in annual revenue by 2050 (\$336 billion by 2035 and \$2.0 trillion by 2040). For comparison, in 2024, the 20 largest global auto makers booked an aggregate of ~\$2.5 trillion in revenue.

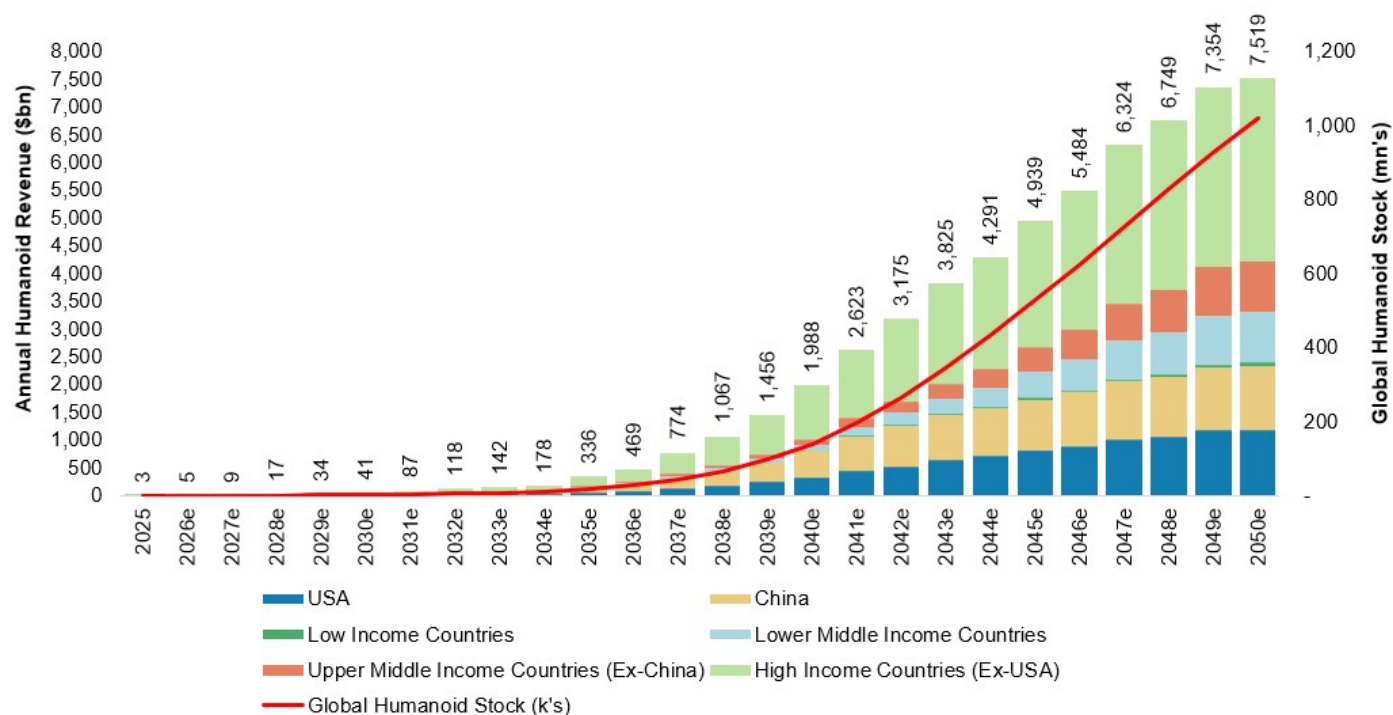
Average Selling Price (ASP) assumptions:

- **High Income Countries:** We assume that the initial ASP of humanoids was \$200k in 2024 (largely research-oriented humanoids), with this declining to \$150k each by 2028, and further declines to ~\$93k by 2040 and ~\$75k by 2050 driven by supply chain development and improvements in mass-production. Assume initial ASPs are inflated given the current lack of domestic supply chains, which get mitigated overtime with onshoring of AI/robotics-enabled manufacturing.
- **Upper Middle/Lower Middle/Low Income Countries:** We assume these humanoids are lower-cost with greater benefits of mass-production, localized supply chains, and production based in cost-efficient countries. Assume meaningful price deflation throughout 2050. We assume an initial ASP of ~\$50k (Rmb350k) in 2024, declining to ~\$21k (~Rmb150k) by 2050.

Exhibit 41: Global Humanoid Market / Revenue Estimates Through 2050 (Assumes 6 Year Replacement Cycle)

Market Size Analysis																			
	2024	2025e	2026e	2027e	2028e	2029e	2030e	2032e	2034e	2036e	2038e	2040e	2042e	2044e	2046e	2048e	2050e		
USA																			
New Humanoid Units Adopted (mn)	-	0.00	0.00	0.01	0.02	0.04	0.065	0.16	0.2	0.7	1.6	3.3	5.0	6.5	7.4	7.7	7.4		
Replacement Units (mn)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.0	0.0	0.1	0.2	0.2	0.7	1.8	3.5	5.8	8.3		
Total Humanoid Units Sold (mn)	-	0.003	0.005	0.008	0.020	0.038	0.065	0.167	0.225	0.722	1.8	3.5	5.8	8.3	10.9	13.4	15.7		
Average Selling Price Per Humanoid (\$k)	200.0	187.5	175.0	162.5	150.0	137.5	130.0	122.3	115.1	108.3	101.9	95.9	90.2	84.9	81.5	78.3	75.2		
USA Revenue (\$bn)	-	0.6	0.9	1.3	2.9	5.3	8.5	20.4	25.9	78.2	182.1	334.7	520.5	702.7	888.5	1,051.1	1,178.9		
China																			
New Humanoid Units Adopted (mn)	0.002	0.012	0.050	0.100	0.176	0.296	0.446	0.923	1.800	4.459	10.262	20.337	31.389	31.9	24.9	15.3	11.5		
Replacement Units (mn)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.0	0.2	0.4	1.0	2.0	4.9	11.2	22.3	36.3	43.1		
Total Humanoid Units Sold (mn)	0.0	0.0	0.0	0.1	0.2	0.3	0.4	1.0	2.0	4.9	11.2	22.3	36.3	43.1	47.2	51.6	54.6		
Average Selling Price Per Humanoid (\$k)	49.0	45.6	38.6	40.5	41.2	37.9	33.4	29.0	25.1	22.1	21.4	20.9	20.5	20.1	20.5	20.9	21.3		
China Revenue (\$bn)	0.1	0.6	1.9	4.0	7.3	11.2	14.9	28.2	49.7	108.3	240.5	465.8	742.6	864.1	965.7	1,077.2	1,161.5		
Low Income Countries																			
New Humanoid Units Adopted (mn)	-	-	-	-	-	-	0.0	0.0	0.0	0.1	0.2	0.3	0.6	1.0	1.5	1.8	1.8		
Replacement Units (mn)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.0	0.0	0.0	0.1	0.2	0.3	0.7	1.2		
Total Humanoid Units Sold (mn)	-	-	-	-	-	-	0.0	0.0	0.0	0.1	0.2	0.3	0.7	1.2	1.8	2.5	3.0		
Average Selling Price Per Humanoid (\$k)	49.0	45.6	38.6	40.5	41.2	37.9	33.4	29.0	25.1	22.1	21.4	20.9	20.5	20.1	20.5	20.9	21.3		
LI Country Revenue (\$bn)	-	-	-	-	-	-	0.1	0.3	1.1	2.2	4.3	7.2	14.1	23.9	37.6	52.3	63.3		
Lower Middle Income Countries																			
New Humanoid Units Adopted (mn)	-	-	-	-	0.0	0.0	0.0	0.2	0.6	1.4	2.6	4.5	8.8	14.5	21.9	26.4	25.7		
Replacement Units (mn)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.0	0.0	0.2	0.6	1.5	2.8	5.1	10.2	17.3		
Total Humanoid Units Sold (mn)	-	-	-	-	0.0	0.0	0.0	0.2	0.6	1.5	2.8	5.1	10.2	17.3	27.0	36.7	43.0		
Average Selling Price Per Humanoid (\$k)	49.0	45.6	38.6	40.5	41.2	37.9	33.4	29.0	25.1	22.1	21.4	20.9	20.5	20.1	20.5	20.9	21.3		
LMI Country Revenue (\$bn)	-	-	-	-	0.8	1.6	1.5	5.1	15.7	32.7	59.2	106.0	209.6	346.7	551.5	764.9	915.0		
Upper Middle Income Countries (Ex-China)																			
New Humanoid Units Adopted (mn)	-	-	-	-	0.0	0.0	0.0	0.2	0.6	1.4	2.6	4.5	8.8	14.6	21.8	26.2	24.6		
Replacement Units (mn)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.0	0.0	0.2	0.6	1.5	2.8	5.1	10.3	17.4		
Total Humanoid Units Sold (mn)	-	-	-	-	0.0	0.0	0.0	0.2	0.6	1.5	2.8	5.1	10.3	17.4	26.9	36.4	42.0		
Average Selling Price Per Humanoid (\$k)	49.0	45.6	38.6	40.5	41.2	37.9	33.4	29.0	25.1	22.1	21.4	20.9	20.5	20.1	20.5	20.9	21.3		
UMI Country Revenue (\$bn)	-	-	-	-	0.8	1.3	1.6	5.3	15.5	32.8	59.9	105.9	209.9	349.7	549.7	760.6	894.6		
High Income Countries (Ex-USA)																			
New Humanoid Units Adopted (mn)	-	0.010	0.015	0.024	0.038	0.104	0.110	0.464	0.571	1.874	4.632	9.487	14.410	18.505	20.466	22.479	20.349		
Replacement Units (mn)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.015	0.038	0.110	0.478	0.609	1.984	5.110	10.096	16.394	23.615		
Total Humanoid Units Sold (mn)	-	0.010	0.015	0.024	0.038	0.104	0.110	0.478	0.609	1.984	5.110	10.096	16.394	23.615	30.562	38.873	43.964		
Average Selling Price Per Humanoid (\$k)	200.0	187.5	175.0	162.5	150.0	137.5	130.0	122.3	115.1	108.3	101.9	95.9	90.2	84.9	81.5	78.3	75.2		
HI Country Revenue (\$bn)	-	1.8	2.6	4.0	5.7	14.2	14.3	58.5	70.1	214.9	520.6	967.9	1,478.7	2,004.2	2,491.0	3,043.0	3,305.2		
Global																			
New Humanoid Units Adopted (mn)	0.0	0.0	0.1	0.1	0.3	0.5	0.7	1.9	3.8	10.0	21.9	42.3	69.0	87.0	97.9	99.9	91.3		
Replacement Units (mn)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.0	0.1	0.3	0.7	2.0	4.1	10.7	23.9	46.4	79.7		
Total Humanoid Units Sold (mn)	0.0	0.03	0.1	0.1	0.3	0.5	0.7	2.0	4.1	10.7	23.9	46.4	79.7	110.9	144.3	179.5	202.2		
Average Selling Price Per Humanoid (\$k)	49.0	119.2	77.0	70.5	64.1	65.3	57.0	59.3	43.4	43.9	44.6	42.8	39.9	38.7	38.0	37.6	37.2		
Global Revenue (\$bn)	0.1	3.0	5.3	9.3	17.4	33.6	40.9	117.7	178.0	469.0	1,066.7	1,987.5	3,175.5	4,291.3	5,484.0	6,749.0	7,518.6		
Global Humanoid Stock (k's)	0	0	0	0	1	1	2	5	11	28	65	138	264	430	624	827	1,019		

Source: Morgan Stanley Research estimates

Exhibit 42: Global Humanoid Revenue by Year (Hardware Sales Only)

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley & Co. LLC and/or Morgan Stanley C.T.V.M. S.A. and/or Morgan Stanley México, Casa de Bolsa, S.A. de C.V. and/or Morgan Stanley Canada Limited and/or Morgan Stanley & Co. International plc and/or Morgan Stanley Europe S.E. and/or RMB Morgan Stanley Proprietary Limited and/or Morgan Stanley MUFG Securities Co., Ltd. and/or Morgan Stanley Capital Group Japan Co., Ltd. and/or Morgan Stanley Asia Limited and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105), Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Daniela M Haigian; Adam Jonas, CFA; Andrew S Percoco; Chelsea Wang; Sheng Zhong.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of July 31, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: Beijing Geekplus Technology Co., Ltd., Estun Automation Co Ltd, Han's Laser, iRay Technology Company Limited, Leader Harmonious Drive Systems, Shenzhen Inovance Technology, Shenzhen SC New Energy Technology Corp, Suzhou Maxwell Technologies Co Ltd, WeiChai Power, Wuxi Autowell Technology Co Ltd, Wuxi Lead Intelligent, Zhejiang Dingli Machinery Co Ltd..

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Zoomlion Heavy Industry.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Shenzhen Inovance Technology, Zoomlion Heavy Industry.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Beijing Geekplus Technology Co., Ltd., China State Construction Engineering, Sany Heavy Industry Co., Ltd., Shenzhen Inovance Technology, Wuxi Lead Intelligent, Zhejiang Hangke Technology, Zoomlion Heavy Industry.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from China State Construction Engineering, CRRC Corp Ltd, Haitian International Holdings Limited, Sinotruk (Hong Kong) Limited.

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Beijing Geekplus Technology Co., Ltd., China State Construction Engineering, Sany Heavy Industry Co., Ltd., Shenzhen Inovance Technology, Wuxi Lead Intelligent, Zhejiang Hangke Technology, Zoomlion Heavy Industry.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: Beijing Geekplus Technology Co., Ltd., China State Construction Engineering, CRRC Corp Ltd, Haitian International Holdings Limited, Sinotruk (Hong Kong) Limited.

An employee, director or consultant of Morgan Stanley is a director of Beijing Geekplus Technology Co., Ltd.. This person is not a research analyst or a member of a research analyst's household. The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all

ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of August 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1555	42%	450	48%	29%	767	44%
Equal-weight/Hold	1577	43%	395	42%	25%	770	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	559	15%	93	10%	17%	216	12%
Total	3,694		939			1754	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O or Over) - The stock's total return is expected to exceed the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Equal-weight (E or Equal) - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.

Not-Rated (NR) - Currently the analyst does not have adequate conviction about the stock's total return relative to the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U or Under) - The stock's total return is expected to be below the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures.

For Morgan Stanley specific disclosures, you may refer to <https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch>.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at <http://www.morganstanley.com/matrix>.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (<http://www.morganstanley.com/terms.html>). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (<http://www.morganstanley.com/terms.html>). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research. Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V. which is regulated by Comisión Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 24-0813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH00001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National

Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of China State Construction Engineering, WeiChai Power, Zoomlion Heavy Industry listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: <http://www.hkex.com.hk>.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

INDUSTRY COVERAGE: China Industrials

COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/31/2026)
Chelsea Wang		
China Railway Group (601390.SS)	E (05/12/2022)	Rmb4.38
China Railway Group (0390.HK)	E (08/11/2025)	HK\$3.45
China State Construction Engineering (601668.SS)	U (08/11/2025)	Rmb4.33
Han's Laser (002008.SZ)	O (10/02/2025)	Rmb96.38
Hefei Meyer Optoelectronic Technology (002690.SZ)	E (09/08/2025)	Rmb16.09
iRay Technology Company Limited (688301.SS)	O (07/07/2026)	Rmb98.62
Neway Valve (Suzhou) Co., Ltd (603699.SS)	O (09/12/2025)	Rmb50.94
Shanghai BOCHU Electronic Technology (688188.SS)	O (08/22/2024)	Rmb99.00
Shenzhen Envicool Technology Co Ltd (002837.SZ)	O (08/19/2024)	Rmb68.21
Sheng Zhong		
Beijing Geekplus Technology Co., Ltd. (2590.HK)	O (08/07/2025)	HK\$9.50
Centre Testing International Group (300012.SZ)	E (11/18/2024)	Rmb15.08
CRRC Corp Ltd (1766.HK)	U (01/22/2026)	HK\$5.34
CRRC Corp Ltd (601766.SS)	U (01/22/2026)	Rmb6.22
DR Laser (300776.SZ)	E (12/17/2021)	Rmb115.58

Estun Automation Co Ltd (002747.SZ)	E (08/25/2026)	Rmb31.66
Haitian International Holdings Limited (1882.HK)	E (09/08/2025)	HK\$18.11
Hongfa Technology Co Ltd (600885.SS)	O (05/23/2023)	Rmb34.65
Jiangsu Guomao Reducer Co Ltd (603915.SS)	U (01/08/2025)	Rmb14.46
Jiangsu Hengli Hydraulic Co.Ltd (601100.SS)	O (05/23/2023)	Rmb102.50
Jingsheng Mechanical & Electrical Co (300316.SZ)	U (01/08/2025)	Rmb43.20
Leader Harmonious Drive Systems (688017.SS)	O (01/22/2026)	Rmb300.49
Sany Heavy Industry Co., Ltd. (600031.SS)	O (01/08/2025)	Rmb20.01
Shenzhen Inovance Technology (300124.SZ)	++	Rmb61.87
Shenzhen SC New Energy Technology Corp (300724.SZ)	U (09/08/2025)	Rmb56.39
Sinotruk (Hong Kong) Limited (3808.HK)	E (05/19/2025)	HK\$40.20
Suzhou Maxwell Technologies Co Ltd (300751.SZ)	U (09/15/2023)	Rmb222.94
Times Electric (3898.HK)	E (01/22/2026)	HK\$31.56
WeiChai Power (2338.HK)	O (03/30/2026)	HK\$31.84
WeiChai Power (000338.SZ)	O (03/30/2026)	Rmb27.17
Wuxi Autowell Technology Co Ltd (688516.SS)	U (09/08/2025)	Rmb45.89
Wuxi Lead Intelligent (300450.SZ)	O (09/08/2025)	Rmb33.62
Zhejiang Dingli Machinery Co Ltd. (603338.SS)	O (11/05/2025)	Rmb57.58
Zhejiang Hangke Technology (688006.SS)	O (09/08/2025)	Rmb21.88
Zhejiang Shuanghuan Driveline Co. Ltd. (002472.SZ)	O (08/25/2023)	Rmb35.51
Zoomlion Heavy Industry (1157.HK)	O (09/08/2025)	HK\$6.10
Zoomlion Heavy Industry (000157.SZ)	O (09/08/2025)	Rmb6.69

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.